

Paediatrics

World of Revision

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Instructions

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NEONATOLOGY : PART 1

----- Active space -----

Neonate : First 4 weeks.

Normal newborn

00:01:00

Routine Newborn Care :

Aseptic precaution :

- Clean hands.
- Clean surface.
- Clean cord cut.
- Clean cord clamp.
- Clean cord.

Early skin to skin contact :

- Avoid hypothermia.
- Establishes breastfeeding.

Others 🐘 :

- Inj. Vit K 1 mg Im (< 1 kg : 0.5 mg) : Anterolateral thigh.
- Eye care.
- Clear secretions.
- Bathing : Not recommended (After cord falls off).

Key updates 2025 :

- Delayed cord clamping : After **atleast 60 sec**
 - Improves iron stores : ↓ anemia
 - Smooth CVS transition
 - Avoids cerebral blood flow fluctuations :
 - ↓ Intraventricular haemorrhage (IVH) : Preterm
- SpO₂ measurement : 2 mins

Weight :

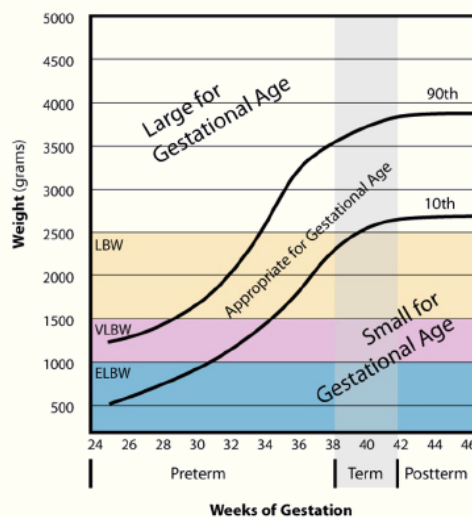
Classification :

Macrosomia : > 4 kg

Low BW : < 2.5 kg

Very low BW : < 1.5 kg

Extremely low BW : < 1 kg



Classification by percentile :

- LGA → • Large for gestational age : > 90th percentile
- AGA → • Appropriate for gestational age : 10th - 90th percentile
- SGA → • Small for gestational age : < 10th percentile

AIIMS Intrauterine growth chart

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Growth abnormalities & Hypoglycemia

00:09:46

Intrauterine Growth Restriction (IUGR) :

SGA + evidence of wasting (Loose skin folds).

Types :

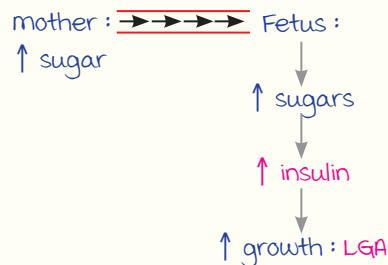
	Asymmetric IUGR	Symmetric IUGR
Cause	maternal : uteroplacental insufficiency	Fetal : Anomalies
Brain	- Brain sparing effect Ⓝ head size, small body	- Involved - Head & Body small
Ponderal index	< 2	> 2

Ponderal Index :

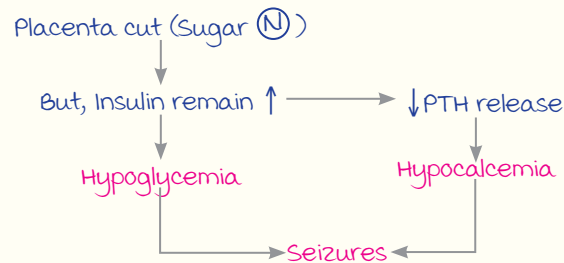
$$PI = \frac{\text{Weight (g)}}{\text{Length (cm)}^3} \times 100$$

Infant of Diabetic mother :

Gestation :



After birth :



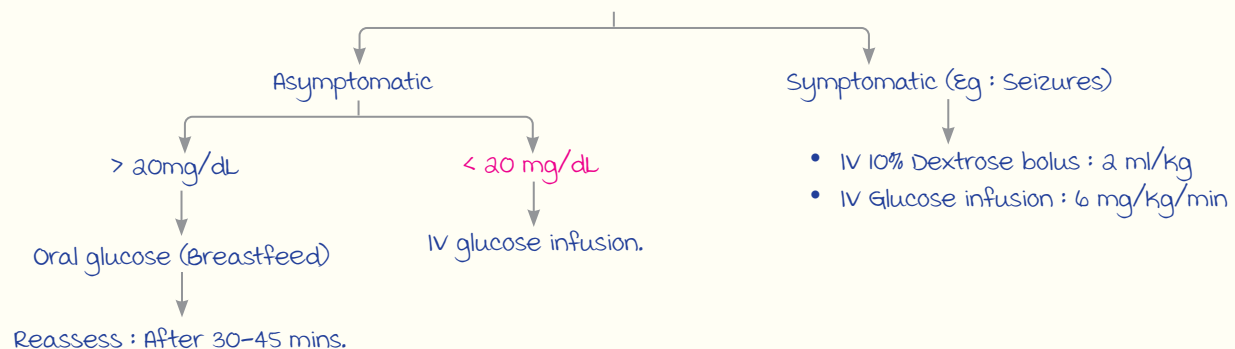
Note :

Growth hormone :

No role → Fetal growth.
→ After birth till 1 year.

management of Hypoglycemia in Newborn :

Hypoglycemia : Blood glucose < 45 mg/dL.

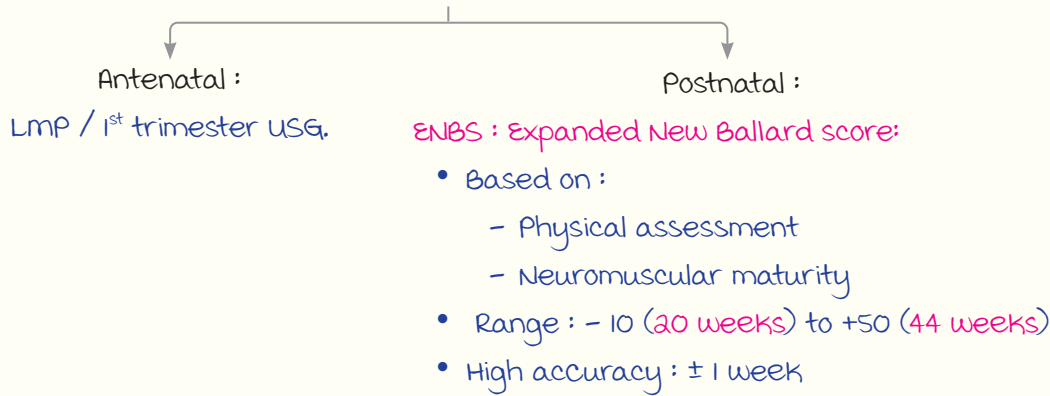


Gestational Age

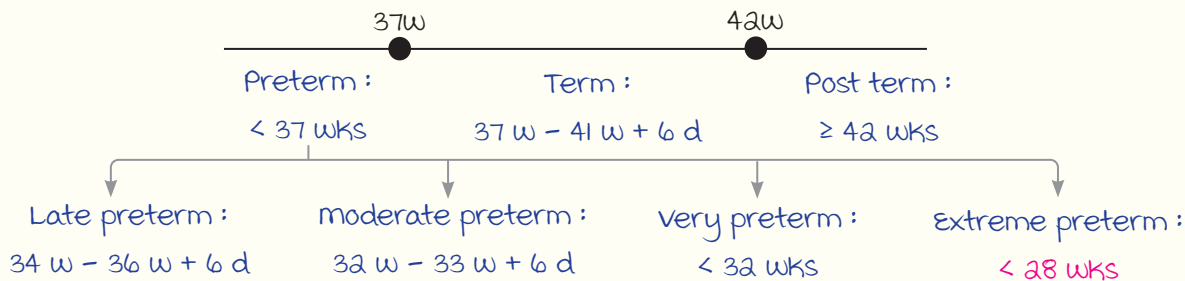
00:21:20

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Assessment :



Classification :



Preterm Baby :

Posture : Extension ($\downarrow$ Tone : Scarf sign +ve).

Breast buds : < 5 mm.

External genitalia :

- Labia majora and minora: Equally visible.
- Scrotum : Less pigmentation, feel empty (undescended).



Scarf sign : Elbow cross midline

Normal Parameters in Newborn

00:29:30

	Normal limit
HR	- Term : 100-160/min - Preterm : 120-180/min
RR	40-60/min
Capillary refill time (CRT)	≤ 3 sec (Shock : $\uparrow$)

----- Active space -----

	Normal limit
meconium 	Passed : ≤ 48 hrs Greenish black (Biliverdin)
urine	Passed : ≤ 24 hrs ↓ Not passed Assess feeding : → Bladder size : → Fluid challenge : ↓ breast feeds (Clinical / USG) : Renal anomalies Obstruction
SpO _a 	- Pulse oximeter : $\textcircled{R}$ upper limb (Preductal blood) - Target SpO_a : 2min : 60-70% 3min : 70-75% 5min : 80-85% 10min : 85-95%
Temperature	$\textcircled{N}$: 36.5 - 37.4 °C - Site : Axilla

Hypothermia

00:38:19

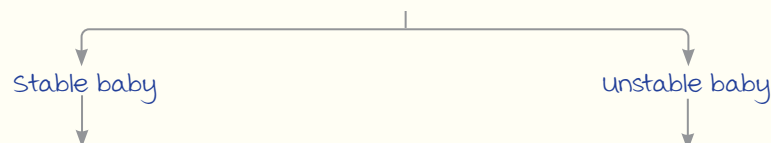
Stages :

	Features	Temp
Cold stress	Extremities cold, body warm	36 - 36.4 °C
moderate hypothermia	entire body cold	32 - 36 °C
Severe hypothermia		< 32 °C

management of Hypothermia :



Frog-leg posture



- Kangaroo mother care (KMC) :
- Skin-skin contact : **Conduction**
 - Prevent hypothermia
 - Posture : Upright, Frog-leg posture
 - Head :
 - max heat loss ($\uparrow$ surface area)
 - Not in contact (Cover separately)
 - Indications :
 - Preterm
 - LBW (especially < 2kg)
 - Duration : min 1 hour

Warmer : mechanism : **Radiation** Incubator : mechanism : **Convection**



Normal Observations in Newborn

00:45:00

----- Active space -----

	Features / Cause	
Erythema Neonatorum 	misomer (Erythema toxicum) • m/c Rash • Red papules + Pustules • Appears > 24 hrs after birth • microscopy : Eosinophils	
milia (milk spots)	whitish spots : Face & nose	Ⓝ inclusion / Retention cysts
Epstein pearls	Pearly white lesions : Palate	
Mongolian spots (Dermal melanocytosis)	Hyperpigmented macules (Greenish-black) : Back	
Small subconjunctival bleeds	d/t stress during delivery	
Stork bite	Benign capillary malformation	

.....
Note :
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Clotting factor XIII deficiency  :

- Delayed bleeding from umbilical stump + Ⓝ PT & APTT.
- Factor XIII : Clot stabilizing factor.
- Test : urea clot solubility test.

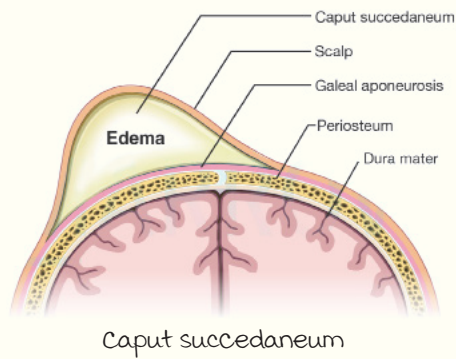
Head Swellings in Newborn

00:54:18

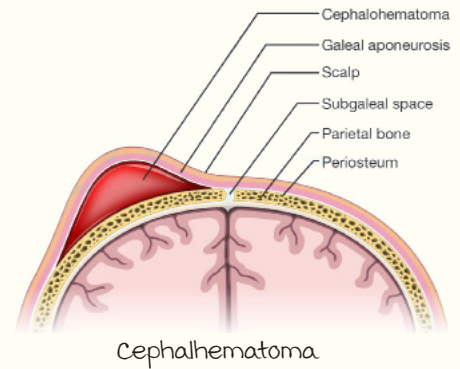
Caput vs. Cephalhematoma :

	Caput succedaneum (m/c)	Cephalhematoma
Cause	Scalp vein congestion d/t prolonged delivery	Trauma d/t instrumental delivery
Content	Fluid	Blood (Subperiosteal area)
Characteristic	Superficial, diffuse	Deep, localised
Appearance	At/Immediately after birth	12-24 hrs after birth
Complications	-	• Jaundice (d/t heme breakdown) • Linear skull fracture

----- Active space -----



Caput succedaneum



Cephalohematoma

Neonatal reflex

00:58:10

moro's reflex :

- Sudden dropping action :
Extension & Abduction $\xrightarrow{f/b}$ Flexion & adduction.



ATNR



moro's reflex

Asymmetric tonic neck reflex (ATNR) :

Head turned to one side :

- 1/L upper & lower limb extension.
- C/L upper & lower limb flexion.

Developmental Timeline of Neonatal Reflexes :

Reflex	Appearance	Fully developed	Duration (PN)
moro 	28 - 32 wks	37 wks	5-6 months
ATNR	35 wks	1 month (PN)	4 months
Parachute (Postnatal reflex)	7-8 month (PN)	-	Persists throughout life
Palmar grasp	28 weeks	-	3 months
Rooting	32 weeks	-	1 month

PN : Postnatal.

Neonatal Resuscitation

01:03:39

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Order of Resuscitation :

T A B C :

Temperature → Airway → Breathing → Circulation.

Steps :

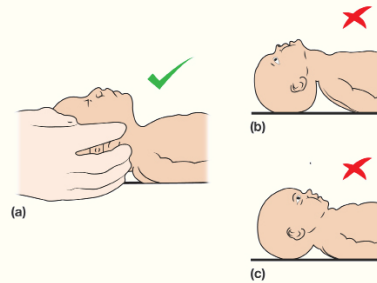
Birth : Assess term, tone, cry

↓ No cry

Start resuscitation

Initial steps :

1. Temperature : Warmer, dry.
2. Airway :
 - Positioning : Slight extension of neck. (Sniffing position).
 - Suctioning of **mouth** → **Nose (if necessary)**.
3. Breathing : Tactile stimulation.
 - Rubbing the back.
 - Flicking the soles.



Sniffing position

No response, check HR

HR < 100 bpm :

Ventilator :

- Route :
 - Bag & mask ventilation
 - **Laryngeal mask**
- Rate : **30-60/min**
- Pulse oximeter compulsory

HR still < 100 bpm

Ventilation corrective steps :

- mnemonic :
 - m : **m**ask reposition.
 - R : **R**eposition head.
 - S : **S**uction of secretions.
 - O : **O**pen mouth.
 - P : **↑P**ressure.
 - A : **A**lternate airway (ET tube)
- **No sequence** to steps.

↓ No improvement :

ET Intubation + Cardiac monitor

↓ If HR < 60 bpm

HR > 100 bpm

(Laboured breathing / Cyanosis) :

- O_a
- CPAP (Preterm)

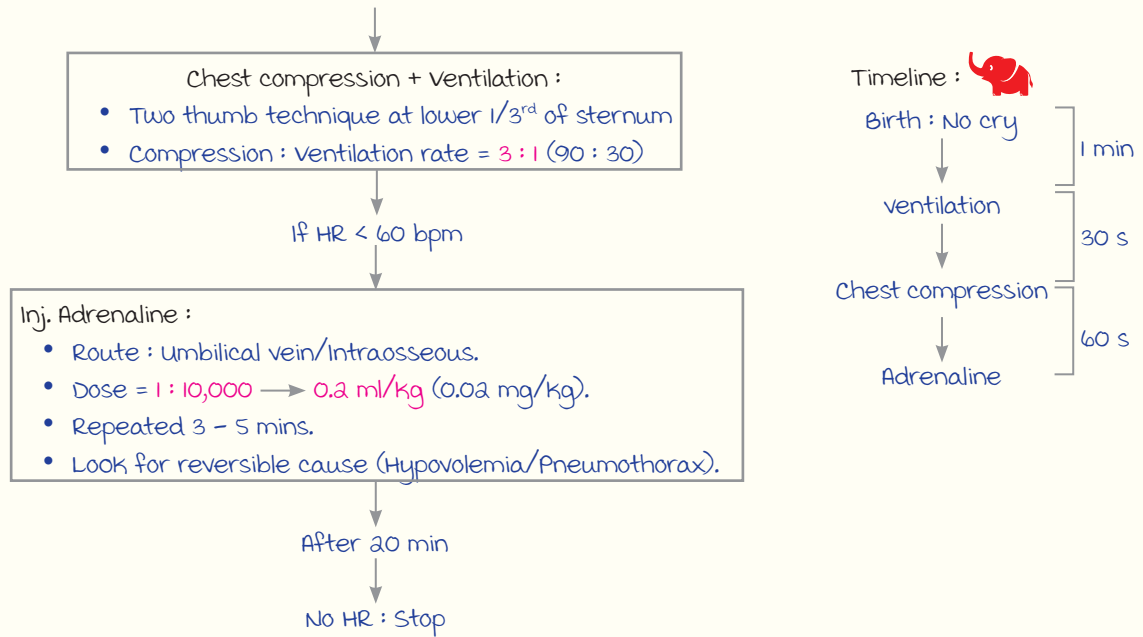
Bag and mask ventilation  :

- Starting O_a concentration :

Gestational age	O_a concentration
≥ 35 wks	21% (Room air)
32-34 wks	21-30%
< 32 wks	30%

- Sensitive indicator of response :
 - HR > 100 /min
 - Chest expansion
- Absolute C/I :
 - Congenital diaphragmatic hernia (CDH).**

----- Active space -----



Resuscitation of a Baby with Meconium Stained Liquor :

- Routine tracheal suction : Not recommended.
- Usual initial steps → Positive pressure ventilation.

NEONATOLOGY : PART 2

----- Active space -----

Fluid Requirement and APGAR Score

00:00:15

Fluid Requirement in Neonates :

Weight (g)	Fluid required (mL/kg/day)				
	Day 1	2	3	4	5
< 1500	80	95	110	120	130
≥ 1500	60	75	90	105	120

APGAR Score :

Assessment of physical condition (1 min and 5 mins after birth).

Components :

	Score		
	0	1	2
Appearance	Cyanosis	Acrocyanosis (Only limbs cyanosed)	Pink
Pulse rate	0	< 100	> 100
Grimace (Reflex irritability)	No response	Grimace	Cry
Activity (Tone)	Flaccid	Partial flexion	Flexion
Respiratory effort	No respiration	Irregular/slow	Cry

Interpretation :

- > 7 : Normal.
- < 7 : Low (Birth asphyxia).

Birth Asphyxia and HIE

00:07:00

Birth Asphyxia/Perinatal Asphyxia :

Injury due to hypoxia/ischemia.

NNF (National neonatology forum) criteria :



----- Active space -----

AAP (American Academy of Pediatrics) criteria  :

- APGAR score : 0 - 3 at > 5 min.
- Umbilical cord pH : < 7 (Acidosis due to anaerobic metabolism).
- Neurological damage :
 - HIE (Hypoxic ischemic encephalopathy) :
 - Hypotonia.
 - Seizures.
 - Coma.
- multiorgan damage.

HIE :

Staging :

	mild	moderate	Severe
Consciousness	Irritable/hyperalert	Depressed	Comatose
Tone	mild hypotonia	marked hypotonic	Flaccid
Seizures	⊖	⊕	⊕ (Prolonged)
Prognosis	Good	• 80% : Good • 20% : Sequelae	• 50% : mortality • 50% : Sequelae

Rx :

- Symptomatic management.
- Induced/therapeutic hypothermia :
 - ↓ Free radical production.
 - Temperature : 33 - 34 °C.
 - used in moderate to severe HIE.

Neonatal Seizures

00:15:45

m/c Cause :HIE (manifests on the 1st day).Associated Electrolyte Imbalances :

- Hypocalcemia.
- Hypomagnesemia.
- Hypo/hyponatremia.

Types :

m/c : Subtle seizures.

Worst prognosis : myoclonic seizures.

Rx :

----- Active space -----

- Phenobarbitone : First drug used in neonates.
- vit. B₆ (Pyridoxine) : used in refractory seizures (↑ GABA → Control seizures).

Seizures in Infant of Diabetic mother :Neonatal Sepsis

00:20:00

Etiology :

Bacterial :

- India : Klebsiella > Acinetobacter.
- Worldwide : Group B streptococcus > E. coli.

Classification :

	Early onset (≤ 72 hours)	Late onset (> 72 hours)
Source	maternal	Nosocomial
Prevention	-	strict hand hygiene

C/f :

Non-specific.

Ix :

Blood sample :

- Culture.
 - Sepsis screening :
 - Positive if any 2 +ve :
 - ↓ TLC : < 5000/mm³.
 - ↓ ANC : < 1800/mm³.
 - ↑ I-T ratio : > 0.2.
 - ↑ CRP : > 1 mg/dL.
 - ↑ micro ESR ≥ 15 mm in 1st hour.
- ANC : Absolute neutrophil count.
 - I : T ratio = Immature : Total neutrophil ratio.

Lumbar puncture : In symptomatic neonate.

mx :

Initial drug : Ampicillin + gentamicin.

meningitis : Ampicillin + cefotaxime.

----- Active space -----

Ophthalmia Neonatorum

00:28:44

Neonatal conjunctivitis.



Ophthalmia neonatorum

Types :

	Gonococci (Severe)	Chlamydia (m/c) 
Onset after birth	2 - 5 days	5 - 14 days
Discharge	Purulent (Copious)	mucopurulent
Complications	- Blindness - meningitis/sepsis	Pneumonia
Rx	Ceftriaxone	macrolides

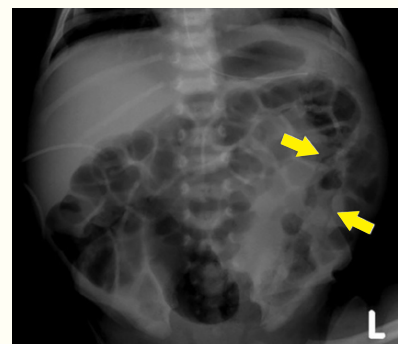
Necrotizing Enterocolitis (NEC)

00:33:06

- Bacterial infection (Intestine) → Necrosis.
- Seen in preterm.

Risk Factors :

- Immaturity of intestine.
- Top feeds (Feeds other than breastmilk).



Pneumatosis intestinalis

modified Bell Staging :

Stages of NEC		Features	
I : Suspected	Ia	Non specific	-
	Ib		+ Blood in stools
II : Definite	IIa	Pneumatosis intestinalis (Air in walls of intestine)	
	IIb	Pneumatosis portalis (Air in portal vein)	
III : Advanced	IIIa	Peritonitis	
	IIIb	Perforation → Pneumoperitoneum	

mx :

- medical (Till IIIa) :
 - NPO + TPN.
 - Broad spectrum antibiotics.
- Surgical (IIIb) : Laparotomy.

Respiratory Distress Syndrome (RDS)

00:40:34

----- Active space -----

- AKA Hyaline membrane disease (HMD).
- Seen in preterm (≤ 34 weeks).

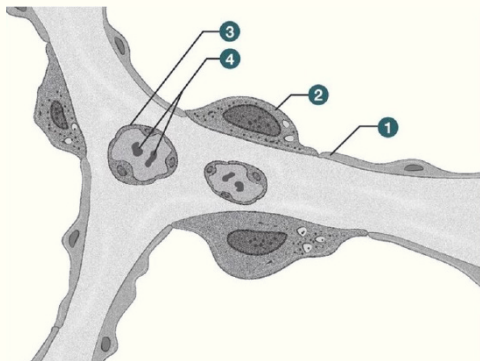
Surfactant 🐘:

Production: —————→ maturity:

- At 18 - 20 weeks of gestation.
 - By type II pneumocytes.
 - > 34 weeks.
 - Lecithin : Sphingomyelin ratio $> 2 : 1$.
- ↓
(DPPC : Dipalmitoyl phosphatidylcholine)

Pathophysiology 🐘:

↓ Immature surfactant → ↑ Surface tension → Alveolar collapse.
↓ Lung compliance.



microscopic image : Alveoli

1. Type I pneumocyte : Flat
2. Type 2 pneumocyte : Elevated and round
3. Capillary
4. RBCs

C/f:

- RR : $> 60/\text{min}$.
- Chest retractions (intercostal).
- Grunting.

Ix:

X-ray:

- White out/ground glass appearance.
- Air bronchogram sign
(Air in bronchus & bronchioles).



Air bronchogram sign

----- Active space ----- Silverman Scoring :

	0	I	2
Upper chest retractions	Synchronized (Chest/abdomen)	Lagging (Chest)	See-saw
Lower chest retractions	Nil	mild	marked
Xiphoid retractions			
Nasal flaring			
Grunt	Absent	mild (Heard with stethoscope)	Severe (Heard without stethoscope)

Interpretation :

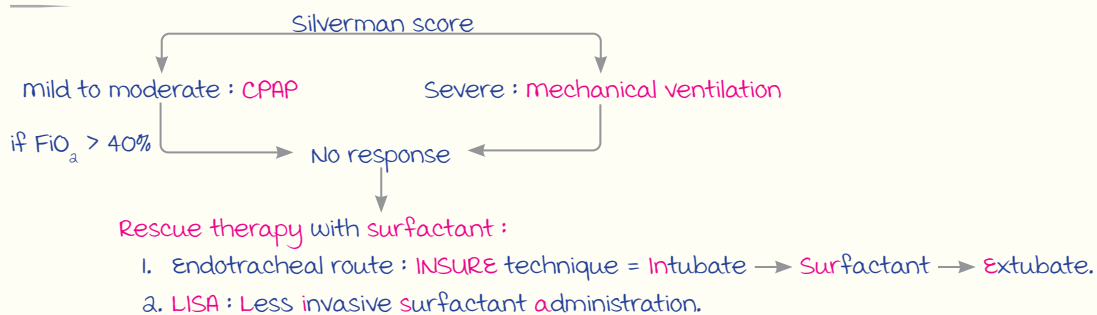
- < 3 : mild.
- $4 - 6$: moderate.
- ≥ 7 : Severe.

Downe Scoring  :

	0	I	2
Respiratory rate	$< 60/\text{min}$	$60 - 80$	> 80
Air entry	Normal	↓	Barely audible
Cyanosis	No	⊕ in room air	⊕ if $\text{FiO}_2 \geq 40\%$
Retractions	No	mild	marked
Grunting	No	mild	Severe

Interpretation : $> 6 \rightarrow$ Impending respiratory failure.

mx :



Prophylactic surfactant : Extreme preterm (< 28 weeks).

Antenatal steroids :

- ↑maturity of surfactant $\rightarrow$ ↓RDS.
- Given at $24 - 34$ weeks of gestation.
- Drugs :
 - Dexamethasone (m/c) :
6 mg BD (4 doses).
 - Betamethasone :
12 mg OD (2 doses).
- Other benefits :
 - ↓NEC.
 - ↓IVH (Intraventricular hemorrhage).
 - ↓Neonatal mortality.

Transient Tachypnea of Newborn (TTNB)

01:02:43

----- Active space -----

- Overall m/c cause of respiratory distress.
- Seen in :
 - Term neonates.
 - Late preterm (35/36th week) neonates.
 - Infant of diabetic mother.
- ↑ Association with LSCS.

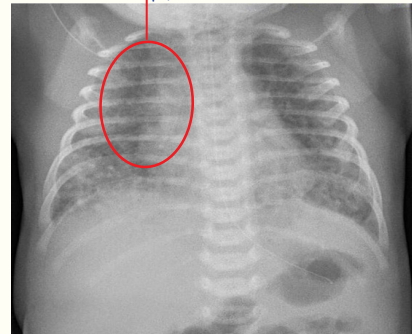
Etiology :

Delayed clearance of lung fluids.

X-ray Features :

- Perihilar lung fluids : 'Sun burst appearance'.
- Fluid in the interlobar fissure.

Sunburst appearance



TTNB

Congenital Diaphragmatic Hernia (CDH)

01:07:02

Herniation of intestines into chest → Compression of lungs → Pulmonary hypoplasia
↓
Respiratory distress.

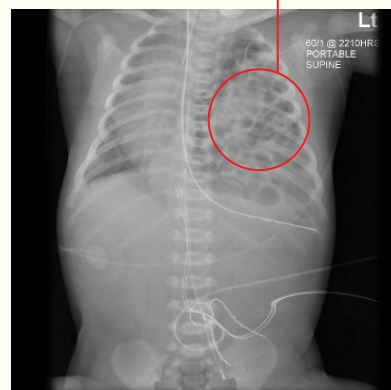
Bochdalek hernia (m/c) :

- Left side (m/c).
- Postero-lateral diaphragm defect.

X-ray Features :

- Intestinal air shadows in thoracic area.
- Absent diaphragm outline.
- mediastinal shift.

Intestinal air bubbles
in thoracic area



CDH

mx :

No cry after birth



1. Resuscitation (Bag and mask : c/d)
- a. Stabilization : mechanical ventilation x 48 hrs



> 48 hours : Surgery.

NEONATOLOGY : PART 3

Neonatal Jaundice

00:00:10

Jaundice :

↑ Bilirubin (N) : ~ 1 mg/dL $\longrightarrow$ Icterus : Bilirubin ≥ 5 mg/dL (Visible jaundice).

Assessment :

Blood test : S. bilirubin.

Screening :

- Transcutaneous bilirubin (TCB) : unreliable in preterm.
- **Kramer's rule** (Cephalocaudal progression of jaundice) :

Area	Approx. bilirubin (mg/dL)
Head (Eyes)	5-7
Chest	7-9
Abdomen	9-11
Limbs	11-13
Palms/soles (Dangerous)	13 - 15 OR more

↓
Cross blood brain barrier $\longrightarrow$ Brain damage.

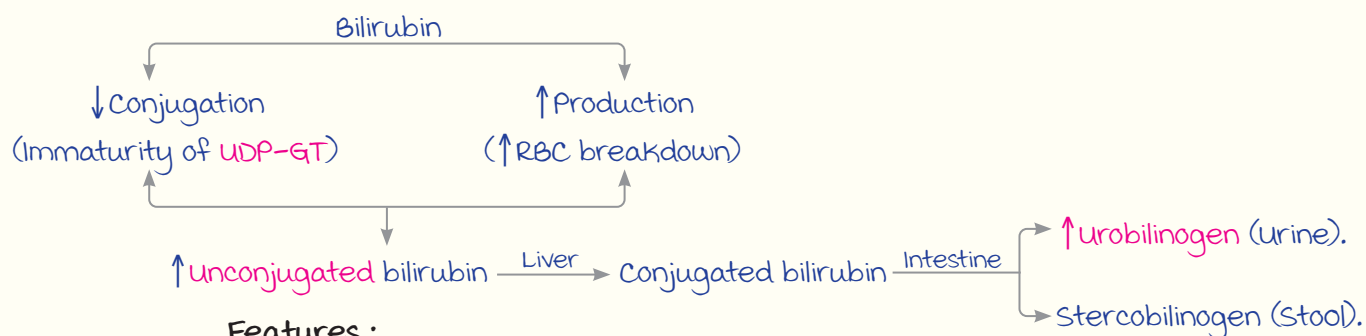


TCB

Physiological Jaundice

00:05:44

Etiology  :



Features :

- Appears > 24 hrs after birth.
- Peaks by 72 hrs.
- Disappearance :
 - Term : 10 - 14 days.
 - Preterm : upto 3 weeks.

Breastfeeding Jaundice and Breast milk Jaundice :

----- Active space -----

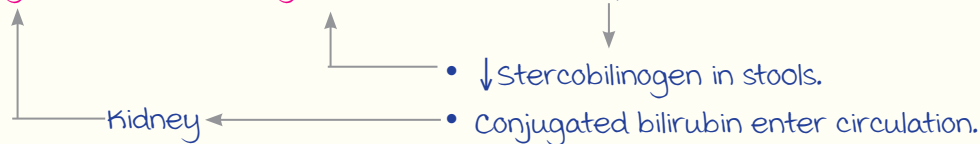
	Breastfeeding jaundice	Breast milk jaundice 🐘
Appearance	1 st week	Beyond 3-4 weeks (Late onset)
Cause	Inadequate breastfeeding ↓ Dehydration : ↓ Bilirubin excretion	• Conj. bilirubin $\xrightarrow{\text{Glucuronidase}}$ Unconj. bilirubin • Free fatty acids : $\ominus$ Conjugation
mx	Adequate breastfeeding	Continue breastfeeding (Self-limiting)

Pathological Jaundice

00:13:56

Features :

- Appearance : Within 24 hrs.
- Duration : >3 weeks.
- Involves palms and soles.
- Dark yellow urine and clay colored stools : Post-hepatic obstruction.

Etiology :

	Cause	Disease
↑ Unconjugated bilirubin (m/c)	↑ Production : Hemolysis	• Inherited : G6PD deficiency, hereditary spherocytosis. • Immunological : Rh/ABO incompatibility
	↓ Conjugation	Crigler-Najjar syndrome : • Type I : UDP-GT $\ominus$ • Type II : Partial deficiency of UDP-GT
↑ Conjugated bilirubin	Neonatal cholestasis	

----- Active space -----

Treatment of ↑ Unconjugated Bilirubin :

Phototherapy 🐘 :

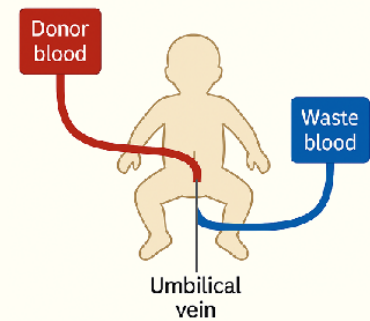
- Plot s. bilirubin on nomogram :
High risk → Phototherapy.
- Blue-green light : 460 – 490 nm.
- Irradiance : 30 $\mu\text{W}/\text{cm}^2/\text{nm}$.
- mechanism :
 - Structural isomerisation
(Bilirubin → Lumirubin : Excreted).
 - Photoisomerisation.
 - Photo-oxidation.



Phototherapy

Exchange transfusion :

- Indications :
 - Very high bilirubin.
 - Signs of bilirubin encephalopathy.
 - Rh isoimmunisation
- Cord blood
 - Bilirubin $\geq 5 \text{ mg/dL}$.
 - Hb $\leq 10 \text{ g/dL}$.



Exchange transfusion

Bilirubin Induced Neurological Damage (BIND) :

Acute (Neonatal) :

- Lethargy.
- Seizures.
- Shrill cry.
- Tone abnormalities :
Hypotonia → Hypertonia
(Opisthotonus).

Chronic (Older children) :

- Cerebral palsy (Basal ganglia) :
- Extrapyramidal symptoms.
 - SNHL.
 - Upward gaze palsy.
 - Enamel dysplasia.



Opisthotonus

Neonatal Cholestasis :

↑ Conjugated bilirubin : $> 1 \text{ mg/dL}$ OR $> 20\%$ of total bilirubin.

Etiology :

Obstruction → Stasis of bile.

----- Active space -----

		Hepatic	Extrahepatic (m/c)
Cause		Neonatal hepatitis : • Idiopathic (2nd m/c) • Secondary : - Infection : TORCH, sepsis - metabolic : Galactosemia, tyrosinemia	EHBA : Extrahepatic biliary atresia (m/c)
Seen in		Preterm/SGA	Term
Appearance		Sick/toxic	Well
Stools		Some pigmentation	Acholic (Clay colored)
Liver biopsy (IHC)	Architecture	Damaged	Ⓝ
	Giant cells	⊕⊕⊕	⊕
	Portal reaction	Lymphocytic infiltration	Extensive fibrosis
	Neo. & periductal proliferation	⊖	⊕

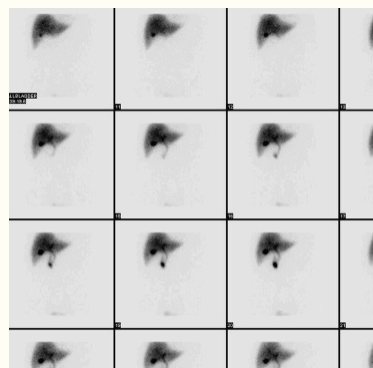
c/f : Dark yellow urine and clay colored stools.

HIDA scan :

- Hepatobiliary iminodiacetic acid scan.
- Tracer :

Tc^{99}
↓
Reach intestine
↓
EHBA ruled out :
High negative predictive value.

- Not diagnostic test.



HIDA scan

mx of EHBA 🐘 :

- **Kasai procedure** : Hepatopertoenterostomy. within 60 days after birth.
- Liver transplantation : Late presentation/liver cirrhosis.

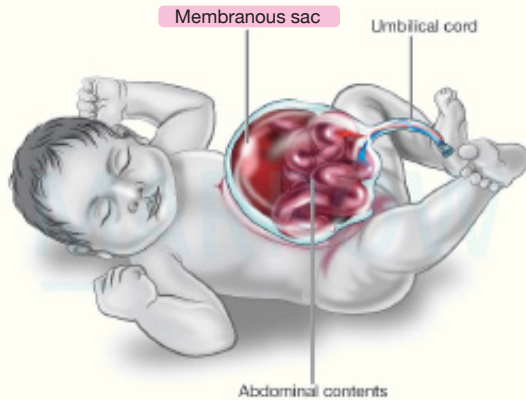
----- Active space -----

Miscellaneous Conditions

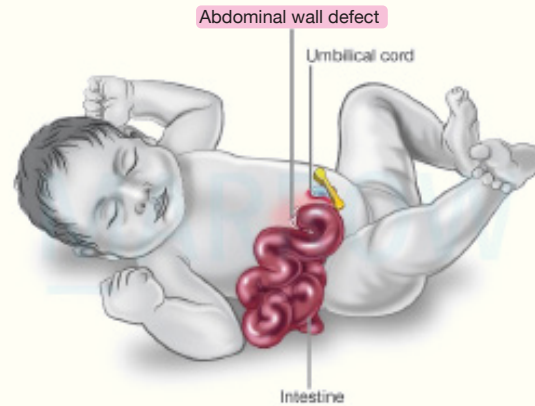
00:40:16

Omphalocele :

- Term baby.
- Intestinal contents covered with sac.
- Umbilical cord **attached to sac**.

**Gastroschisis :**

- Preterm.
- Abdominal wall defect : No sac.
- Umbilical cord **adjacent to intestine**.

**Cloacal exstrophy :**

1. **Ectopia vesicae** : Bladder exstrophy.
2. Terminal ileum.
3. Omphalocele.

**Amniotic band syndrome 🐘 :**

Fibrous bands of amnion compresses limbs



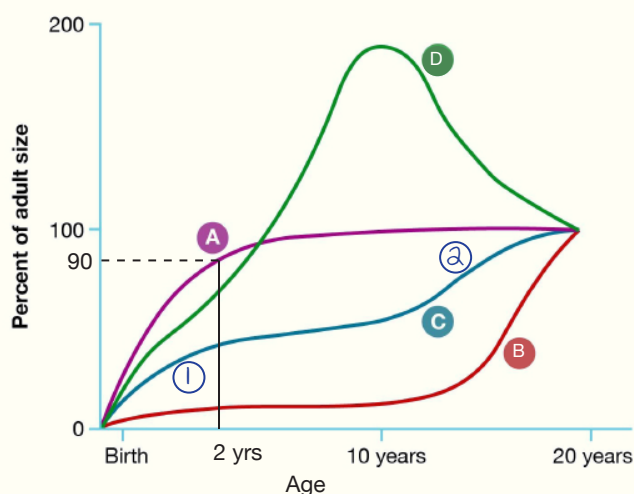
Constriction rings : Limb defects, amputation.



GENERAL PAEDIATRICS : GROWTH

----- Active space -----

Scammon's Growth Curve :



Growths :

- A** : Brain (max : 2 yrs)
- B** : Gonads
- C** : Somatic (1, 2 → Growth spurts)
- D** : Lymphoid tissue

Anthropometry

00:04:12

Best indicator of growth : **Consistent** ↑ in height and weight.

Weight :

Average birth weight : 3 kg.

Weight loss after birth :

	Weight loss % (In 1 st week)	Regains weight by
Term	10	10 th day
Preterm	15	15 th day

Weight gain  :

Age	↑ in weight
Till 3 months	20 - 40 g/day
Till 1 yr	400 g/month
1 - 7 yrs	2 kg/yr
> 7 yrs	3 kg/yr

- Doubles : 5 months
 - Triples : 1 year
 - Quadruples : 2 years
- } After birth.

Height/Length :

measurement :

- Infantometer : Length (< 2yrs).
- Stadiometer : Height.

----- Active space -----

Increase in height 🐘 :

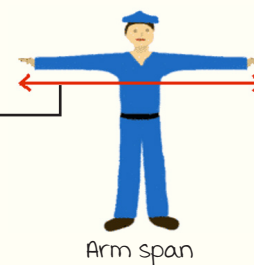
Age	Height
At birth	50 cm
At 1 year	75 cm
At 2 years	87.5 cm
> 2 yrs - Till puberty	↑ of 6 cm/yr
Growth spurt at puberty	♂ : 20 - 30 cm ♀ : 16 - 28 cm
Adult	160 - 170 cm (Avg)

- Half of adult height (80 - 85 cm) : At 20 - 24 months.
- Doubles (100 cm) : 4 years.
- Triples (150 cm) : 12 years.

Arm span 🐘 : Height equivalent in older children.

Comparison	Age
Arm span (~2.5 cm) < Length	Birth
Arm span = Height	11 yrs
Arm span (> 1 cm) > Height	> 11 yrs

Tip of middle finger from one hand to another

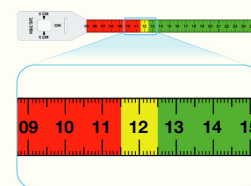
Circumference :

mid arm circumference (MAC) :

- Nutritional status assessment.
- Age : 1 - 5 years.

Shakir's tape :

- Normal : > 12.5 cm.
- malnutrition : 11.5 - 12.5 cm.
- Severe malnutrition : < 11.5 cm.



Shakir's tape

Head circumference (HC) :

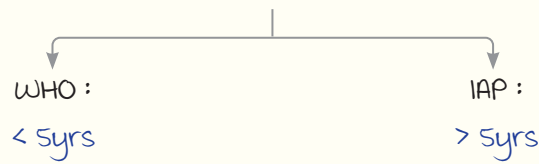
- AKA Occipito-frontal circumference (OFC).
- Brain growth assessment.

Age	Head circumference & growth rate
At birth	33 - 35 cm (34 cm)
First 3 months	2 cm/month (40 cm)
3 - 6 months	1 cm/month (43 cm)
6 months - 1 yr	0.5 cm/month (46 cm)
1 - 2 yr	2 cm/year (48 cm)
At 12 yrs	52 cm

Growth Charts

00:18:06

----- Active space -----



WHO Growth Chart (2006) :

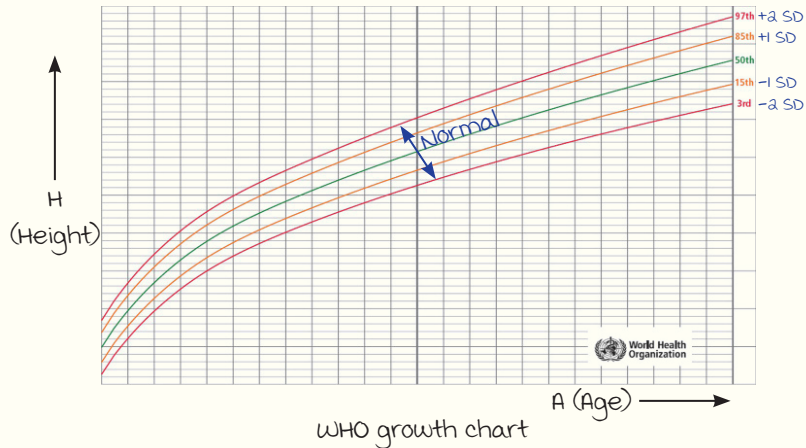
Based on multicentric growth reference study (MGRS).

Colors :

- Blue : Boys.
- Pink : Girls.

General interpretation :

	Low	High
Percentile	< 3rd	> 97th
S.D.	< -2	> +2



Puberty

00:25:07

Female (8 - 13 years) :

Thelarche (Breast enlargement)

↓
Pubarche (Pubic hair appearance)

↓ Growth spurt (GS)
menarche.

male (9 - 14 years) :

Increase in size of testes

↓
Penile enlargement

↓
Pubarche

↓ GS
Facial/Axillary hair
(Later part : voice change)

Tanner's Staging (Sexual maturity Rating) :

Stages	♀		♂	
	Breast	Pubic hair	Testes/ Penis/ Scrotum	
I	Prepubertal (No breast buds)	No terminal hair	Prepubertal	
II	Breast buds appear	Sparse	Testicular volume : 4 mL	
III (♀ GS)	Swelling of breast	Thick curly	Penile enlargement	
IV (♂ GS)	2° mound (Nipple areola swelling)	Coarse	Dark scrotum	
V	Adult - like	Adult - like (Extends to medial thigh)	Adult - like	

----- Active space -----

Short Stature

00:34:18

Height for age (WHO chart) : < 3rd percentile (or) -2 SD.

Causes :

Normal > Pathological :

	Constitutional delay (m/c)	Familial short stature
Adult height	Normal	Short
Puberty	Delayed	Normal
Bone age		

measurements :

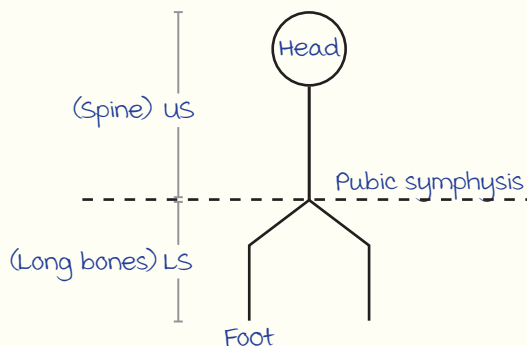
mid parental height (Prediction) :

$$\text{MPH} = \frac{\text{Father's height} + \text{mother's height}}{2} \pm 6.5 \text{ cm}$$

- Boys : + 6.5
- Girls : - 6.5

US : LS Ratio :

Age	US : LS
At birth	1.7 : 1
3 yrs	1.3 : 1
10 yrs	1 : 1
> 10 yrs	0.9 : 1



US : upper segment
LS : Lower segment

Approach to Short Stature :

US : LS ratio		
Normal : Proportionate	Abnormal : Disproportionate	
	Short trunk/US	Short limbs/LS
• Normal variants • Chronic malnutrition • GH deficiency (Short : > 1yr)	vertebral defects (E.g. : Spondylo-epiphyseal dysplasia)	• Achondroplasia • Rickets • Congenital hypothyroidism

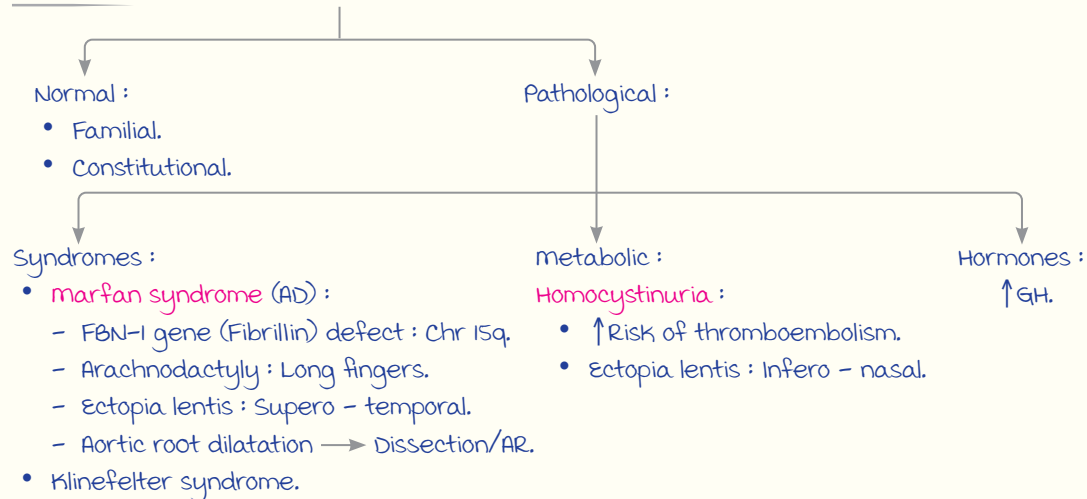
Tall Stature

00:42:26

----- Active space -----

Height for age (WHO chart) : $> 97^{\text{th}}$ percentile (or) $+ 2 \text{ SD}$.

Causes :



marfans syndrome :
Kyphoscoliosis ⊕

Microcephaly

00:47:09

Head circumference for age $< -2 \text{ SD}$.

Causes :

- 1°/Genetic : Trisomies.
- 2°:
 - Prenatal : Teratogens.
 - Perinatal : Asphyxia.
 - Postnatal : Trauma & infections.

Cri-du-chat syndrome :

- Partial deletion : Chr 5p(-).
- Cat like cry (D/t tracheal hypoplasia).

Rubinstein Taybi/Broad thumb syndrome :

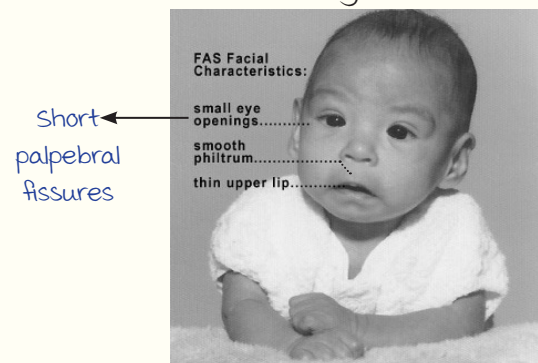


Broad toe

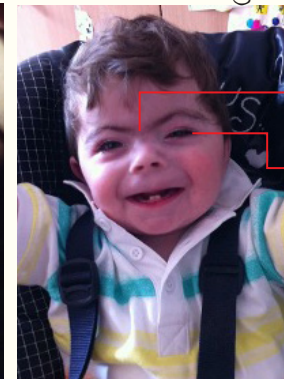


Broad thumb

Fetal alcohol syndrome 🐘 :



Cornelia-de Lange syndrome :



Synophrys/
unibrow
Long eye lashes

----- Active space -----

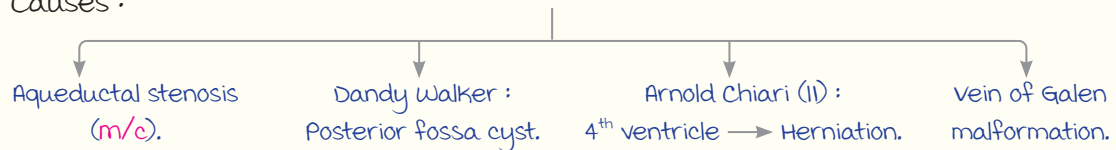
Rett's syndrome :

- X-linked dominant.
 - Defect : **MECP2 defect**.
 - Clinical features :
 - microcephaly.
 - Developmental **regression**.
 - **Stereotypes** : Hand wringing movements.
 - Speech defects, ataxia.
 - Rx : Trofinetide (Analog of IGF-I).
- Normal 1 yr (At birth) → Symptoms ⊕.

Congenital Hydrocephalus

00:55:18

Causes :



c/f :

1. macrocephaly (D/t ↑ CSF).
2. Bulging fontanelle.
3. Prominent scalp veins.
4. **Sunset appearance** : visible upper sclera.
5. **Cracked pot resonance** (D/t ↑ pressure) : **macewen's sign**.



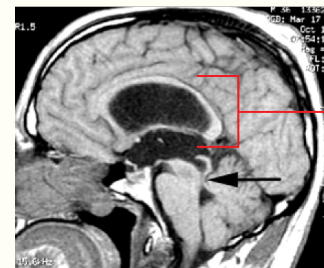
Features of macrocephaly

Aqueductal Stenosis :

- B/w 3rd and 4th ventricle.
- Narrowing/obstruction

↓
 ↑ Size : Lateral and 3rd ventricle. } (m/c cause of fetal ventriculomegaly)

mx : Ventriculo - peritoneal shunt (V-P Shunt).

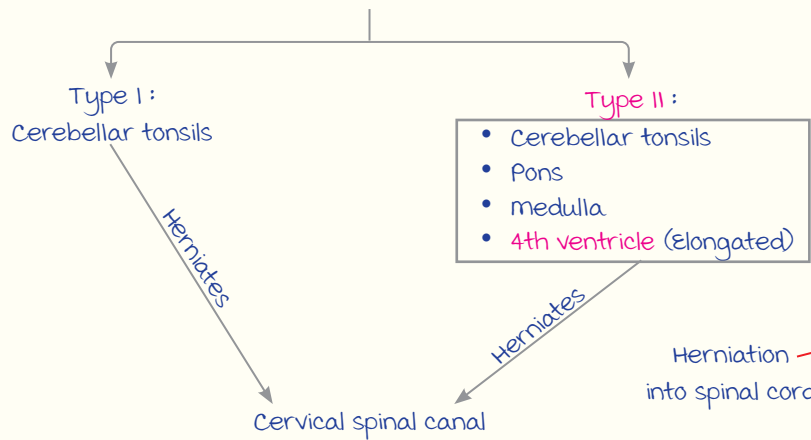


Aqueductal stenosis

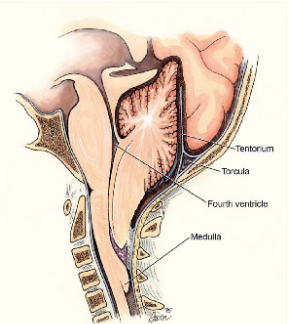
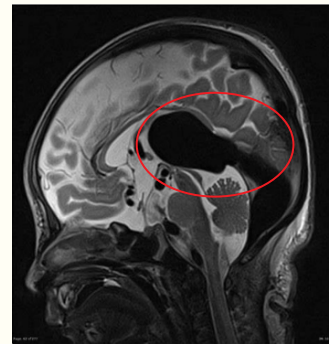
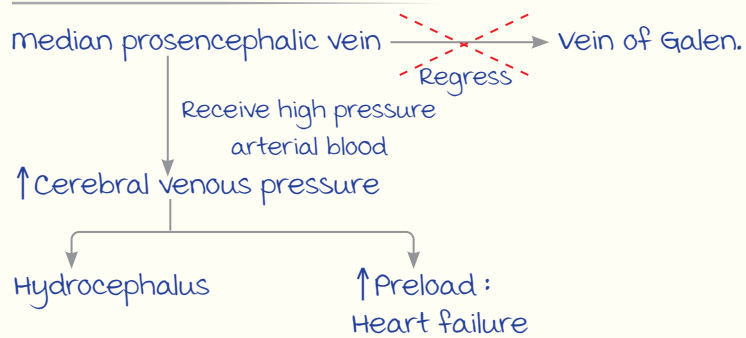
↑ Size
of 3rd &
lateral
ventricle

Arnold Chiari malformation :

----- Active space -----



Arnold Chiari malformation type 2

vein of Galen malformation :

vein of Galen malformation

----- Active space -----

DEVELOPMENT

4 domains of development :

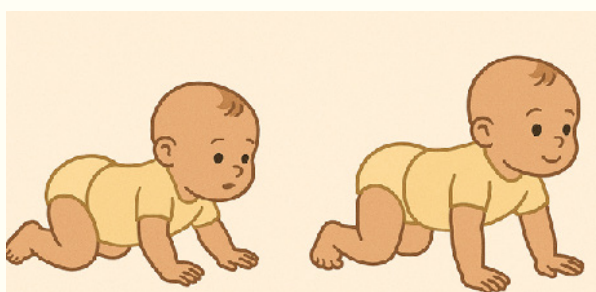
1. Gross motor.
2. Fine motor.
3. Language.
4. Social.

Gross Motor Milestones

00:02:27

Cephalo-caudal.

Age	milestone attained
3 months	Neck holding
4 months	Rolling over
6 months	Sitting with support, tripod posture
8 months	Sitting without support, crawling
10 months	Standing with support, creeping
12 months	• Standing without support • walking with support
15 months 	• Walking independently • Creeping upstairs
18 months	Running
2 years	Climbing stairs with 2 foot/step
3 years	• Climbing upstairs with 1 foot/step • Rides tricycle
4 years	• Climbing downstairs with 1 foot/step • Hops
5 years	Skiping



Crawling
(8 months)

Creeping
(10 months)

Fine Motor Milestones

00:10:51

----- Active space -----

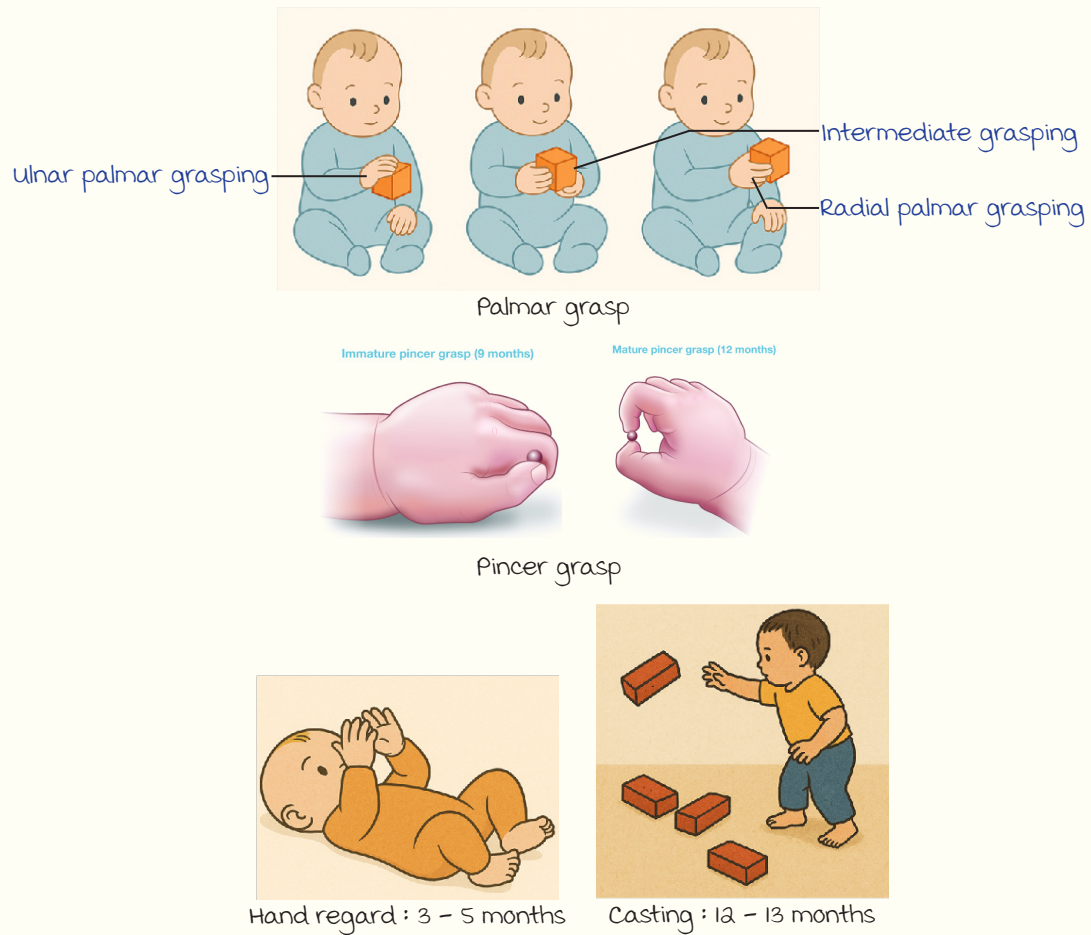
Proximal (Palm) to distal (Fingers).

Age	milestone attained
4 months 	• Bidextrous grasp + Bringing hands to midline - Reaching out to objects • mouthing • Hand regard (3 - 5 months)
6 months 	• unidextrous grasping • Transferring of objects - Signifies the ability to compare objects • ulnar palmar grasping (Immature)
8 - 9 months	Intermediate palmar grasping
9 - 10 months	Immature pincer grasping
10 months	Radial palmar grasping (mature)
12 months	• mature pincer grasping • Casting (Deliberately throwing objects)
15 months	• Imitates scribbling • makes tower of 2 cubes
18 months	• Scribbles well • makes tower of 3 cubes • Starts feeding with spoon (Spills)
2 years	• Draws line • makes tower of 6 cubes • Feeds well with spoon
3 years	• Draws circle • makes tower of 9 cubes
4 years	Draws cross
4.5 years	Draws square
5 years	Draws triangle
7 years	Draws diamond/rhomboid

mnemonic for copying shapes : '**LOX STD**'.

- 2 years : **L**ine.
- 3 years : **O** (Circle).
- 4 years : **X** (Cross).
- 4 ½ years : **S**quare.
- 5 years : **T**riangle.
- 7 years : **D**iamond/rhomboid.

----- Active space -----



Language Milestones

00:23:00

Sounds → Syllables → Words.

Age	milestones attained
3 months	Cooing (musical) sounds
4 months	Laughing aloud
6 months	monosyllables
9 months	Bisyllables
1 year 	1 - 2 words
18 months 	8 - 10 words
2 years 	• 50 - 100 words • 2 - 3 word sentences
3 years	• Can state name, age, gender • Asks questions
4 years	Can tell stories and rhymes
5 years	Asks meaning of words

Social Milestones

00:27:35

----- Active space -----

Age	milestone attained
2 months	Social smile
3 months	mother regard
6 months	- Stranger anxiety - Smiles at mirror image
9 months 	- Waves 'bye-bye' - Plays peek-a-boo
1 yr 	- Plays simple ball game - Comes when called
15 months	- Jargon speech - Points to objects
18 months	Domestic mimicry
2 yrs	- Starts toilet training - Asks for food - Listens to stories
3 yrs	Shares toys
4 yrs	- Goes to toilet alone - Group play - Right-left discrimination
5 yrs 	Dresses and undresses without help

Special Milestones

00:34:42

Handedness  :

- Predominance of one hand over the other.
- Attained at 3 years.

Object permanence/constancy  :

- Understanding that objects continue to exist even when they are not seen.
- Attained at 9 months.



Pivoting (10 - 11 months)



Object permanence



Pretend play (4 years)

----- Active space -----

Red Flag Signs and Developmental Delay

00:38:06

Red Flag Signs :

milestone	upper limit (Age)
Vocalization	6 months
Sitting	10 months
Standing	17 months
Walking	18 months
Single words	18 months

Developmental Delay :

Developmental quotient (DQ) :

- $\frac{\text{Developmental age}}{\text{Chronological age}} \times 100$
- (N) : $> 70\%$.
- Developmental delay : $< 70\%$.

Global developmental delay :

- $DQ < 70\%$ in ≥ 2 domains.
- E.g. : Cerebral palsy.

Nocturnal Enuresis

00:41:12

- Bed-wetting ≥ 2 times a month.
- In children > 5 years of age.
- usually 1^o nocturnal type (From birth till evaluation).

Cause : Delay in bladder maturation (m/c) $\rightarrow$ Normal variant.

management :

Non-pharmacological :



Pharmacological : If unresponsive to non-pharmacological mx.

- Desmopressin :
 - Oral drug.
 - Administered 2 hours before bedtime.
 - Synthetic vasopressin (ADH).
 - mimics the (N) circadian rhythm of (ADH).
- Oxybutynin (Anticholinergic) : ↓ Bladder contraction.

GENERAL PAEDIATRICS : NUTRITION

----- Active space -----

Breast Feeding

00:00:15

Breastfeeds :

Initiation in newborn : As soon as possible/within 1 hr of birth.

Composition :

Secretion	Time since birth
Colostrum	3 - 4 days
Transitional milk	Till 14 days
mature milk	> 14 days

Colostrum  :

- Thick, yellow.
- Quantity : Less.
- Immunological components :
 - IgA (most abundant Ig).
 - Anti-infective proteins.
 - Vitamins : A, D, E, K.

Latching :

Signs of good attachment :

1. Wide open mouth.
2. Everted lips.
3. Chin touching mother's breast.
4. Areola visibility : upper > Lower.



Nutritional Properties :

Nutritive value : 67 Kcal/100 mL.

Breast milk : For overall neurodevelopment.

Composition :

	w.r.t cow milk	major components
Carbohydrate	↑	Lactose : ↓ Lactoferrin (Antibacterial) Glucose + Galactose → Galactocerebroside : myelination
Protein	↓	• Lactalbumin } easily digestible • Lactoglobulin } • Taurine } (+) Brain growth, (-) in cow milk • Cysteine }
Fats	Same	PUFAs : • Docosahexaenoic acid (DHA) } (+) Brain growth • Arachidonic acid (ARA) }

Exclusive breast feeding (EBF) : Till 6 months.

Complementary feeding (Breast milk + other feeds) : 6 months - 2 years
(To promote brain growth).

----- Active space -----

Immunological Properties :

- Lactoferrin.
 - IgA.
 - Bifidus factor $\oplus$ Growth $\rightarrow$ Bifidobacterium.
- } $\downarrow$ Risk of infections (RS/GIT).

Storage of Expressed Breast milk (EBM) :

Room temperature : 6 - 8 hrs.

Refrigeration : 24 hrs.

Freezer (-20°C) : 3 months.

Best alternative to breast milk : Donor human milk.

Feeding in Preterm :

Gestational age (wks)	Developmental features	mode of feeds
< 28/hemo-dynamically unstable	$\downarrow$ Gut motility & function	TPN
28 - 31	Incoordinated swallow & respiration ($\uparrow$ Risk of aspiration)	NG/OG tube
32 - 34	Immature sucking reflex	EBM : Paladai/katori spoon
> 34	Swallow $\oplus$, sucking reflex $\oplus$	Direct breast feeds

Complementary Feeding :

Consistency :

Semisolid/thick liquid $\xrightarrow{\text{F/b}}$ Solids.

Age	Quantity of feeds	Frequency of feeds
6 m - 1 yr	$\frac{1}{2}$ - 1 katori	3 times/day
> 1 yr	1 - $1\frac{1}{2}$ katori	5 times/day

Absolute C/I of Breast Feeds :Infant  :

metabolic :

- Lactose intolerance.
- Galactosemia.

maternal : Chemo/radiotherapy.

HIV x Breast feeding :

- Relative C/I for BF.
- Replace with formula milk if available.
- Feeds : EBF for 6 months
 - $\downarrow$
 - Continue breastfeeds for 1 yr
 - $\downarrow$
 - Gradual cessation over a month.

Antiretroviral prophylaxis :

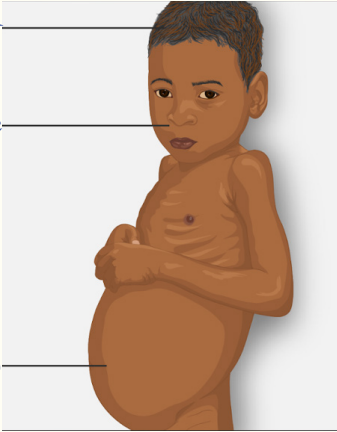
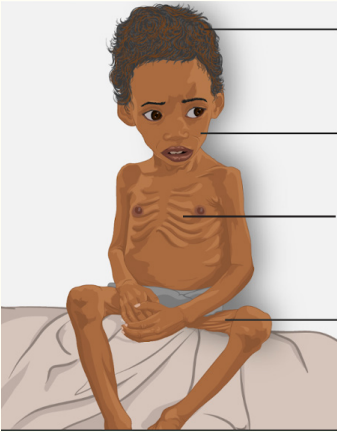
Risk stratification	Prophylaxis
High risk : • mother not on ART • High viral load (> 1000 copies/mL) • Recent Dx (< 6 wks postpartum)	Nevirapine + Zidovudine x 12 wks
Low risk : mother on ART	Nevirapine x 6 wks

Kwashiorkor & Marasmus

00:25:38

----- Active space -----

Protein energy malnutrition.

	Kwashiorkor	Marasmus
Age at onset	> 1 yr	< 1 yr
Predominant deficiency	Protein (Albumin)	Calories
Characteristic features	Edema (+) (Anasarca)  Thin & brittle hair moonface Bulging abdomen (Ascites)	Wasting of fat & muscles (Edema (-))  Thin & brittle hair Simian/old man facies Skin & bone appearance (Prominent rib borders) Loose skin folds
Appearance	Dull	Alert
Appetite	↓↓	Intact
Prognosis	Poor	Good

Additional features of Kwashiorkor :

- Fatty liver (D/t deficiency of fat transporting proteins).
- Flaky paint appearance.
- Crazy pavement dermatosis : Cracks on skin.
- Flag sign : Alternative black & white pigmented hair.



Flaky paint appearance

WHO Classification of malnutrition

Acute malnutrition :

- Wasting.
- ↓Weight for height :
 - moderate : -2 to -3 SD.
 - Severe : < -3 SD.

Chronic malnutrition :

- Stunting
- ↓Height for age :
 - moderate : -2 to -3 SD.
 - Severe : < -3 SD.

Edematous malnutrition : If edema (+)

----- Active space -----

Severe Acute Malnutrition (SAM)

00:37:45

Criteria :

Any 1 ⊕ :

- weight for height < -3 SD.
- B/L pedal edema.
- mid arm circumference < 11.5 cm (Red zone on Shakir's tape) : 6 months to 5 yrs.

Assessment :

	uncomplicated SAM	Complicated SAM
Appetite	Good	Poor
Complications	⊖	⊕
Edema	minimal/⊖	Significant ⊕
Prognosis	Favourable	Unfavourable
Rx	Supervised home care	Hospital mx

management & Complications :

Phases		
Stabilization (1st week) : mx of complications. Rehabilitation (> 1 wk) : Restore nutrition.		
Complications	Features	Treatment 
Sugars (Hypoglycemia)	Blood glucose < 54 mg/dL	• Symptomatic : IV 10% dextrose 5 mL/kg • Asymptomatic : Oral feeds
Hypothermia	Axillary temperature $< 35^{\circ}\text{C}$	-
Infections	Gram -ve Enterobacteriaceae (m/c)	IV Ampicillin + Gentamicin
Electrolyte disturbances	$\uparrow$ Total Na^+ , $\downarrow$ K^+ , $\downarrow$ Mg^{2+}	SAM + Shock : RL + 5% dextrose
Dehydration	D/t associated diarrhea	ReSoma1
Deficiencies	micronutrients	1 st wk avoid iron Rx : D/t complications $\rightarrow$ $\downarrow$ Absorption $\rightarrow$ $\uparrow$ Free radicals $\rightarrow$ Damage intestine

Rehydration solution for malnourished children (ReSoma1) :

Characteristics	w.r.t WHO ORS (245 mmol/L)
Osmolarity	300 mmol/L
Glucose	$\uparrow$
micronutrients (Zn, Cu, Mg)	⊕
Na^+	$\downarrow$
K^+	$\uparrow$

Feeding of child with SAM :

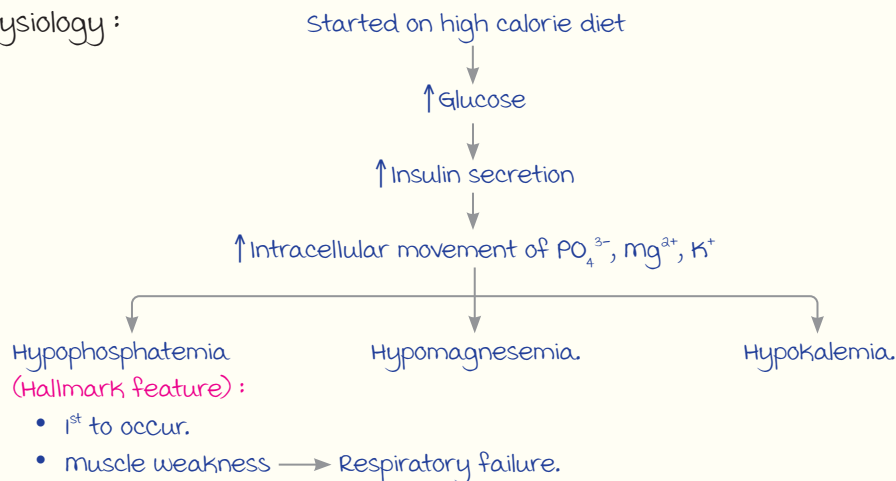
----- Active space -----

1st week : F75 $\xrightarrow{\text{Increase gradually}}$ 2nd week : F100

- (75 kcal + 0.9 g protein)/100 mL.
- milk based.
- Low calorie (Starter feeds).
- (100 kcal + 2.9 g protein)/100 mL.
- Nutritionally dense.

Refeeding Syndrome

Pathophysiology :



Failed Response to Treatment :

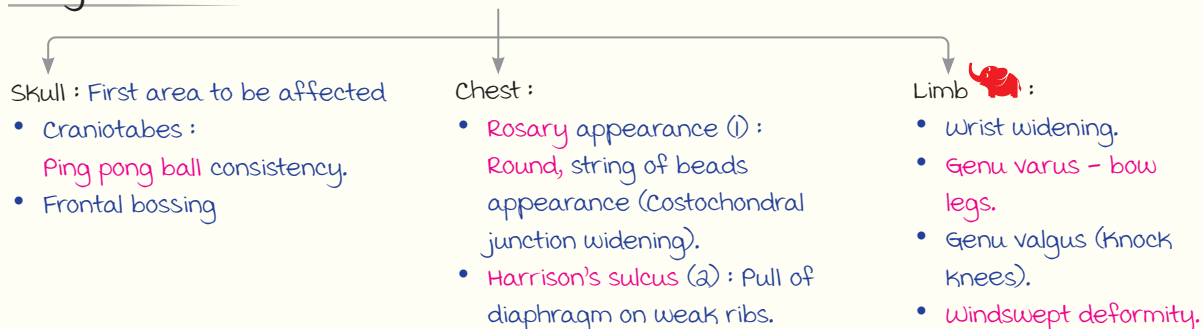
	Primary failure 	Secondary failure
Clinical response	Absent since initiation of Rx	Initially ⊕ $\xrightarrow{F/b}$ Failure to respond later
Criteria	Failure to : • Regain appetite by day 4 • Start losing edema by day 4 • Completely lose edema by day 10 • ↑ weight by 5 g/kg/day by day 10	Failure to : Gain weight atleast 5 g/kg/day for 3 consecutive days
Etiology	untreated/inadequately treated infection	Improper feeding

Rickets

01:00:43

vit D deficiency $\rightarrow$ ↓ mineralisation of bone (m/c : metaphysis).

Early Features :



----- Active space -----



Rosary appearance,
Harrison's sulcus



Wrist widening



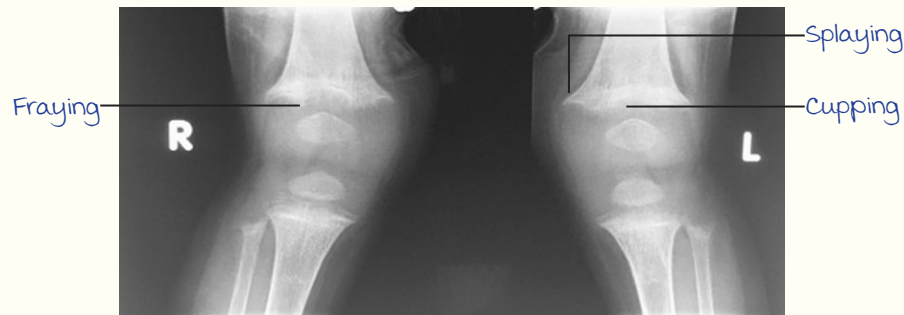
Genu varus



Genu valgus

Investigation :

X-ray 🐘 : First lx.



Biochemical lx :

- Ca^{2+} : ↓/Ⓝ.
- PO_4^{3-} : ↓.
- PTH : ↑.
- ALP : ↑.

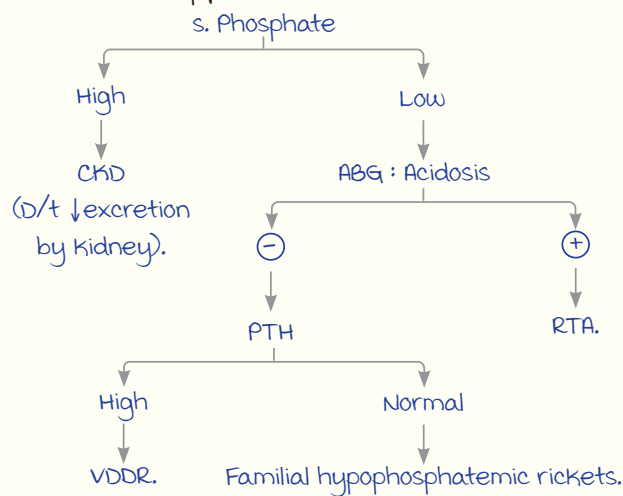
Rx :

- Vit D : 3 - 6 lakh units.
 - Divided dose therapy.
- Not responding : Refractory rickets.

Refractory Rickets 🐘 :

		Features
Vitamin D dependent rickets (VDDR)		• Autosomal recessive • Type 1 : 1α hydroxylase deficiency → ↓$1, 25 - (OH)_2 D_3$ • Type 2 : Vit D resistance (Receptor defect) → ↑$1, 25 - (OH)_2 D_3$
Familial hypophosphatemic rickets		• X-linked dominant • PHEX gene → Endopeptidase → maintains PO_4^{3-} • PHEX gene → mutation → Hypophosphatemia → Rickets
Others	CKD	Late CKD (GFR < 25%) → ↓ $1-\alpha$ hydroxylase activity → ↓Active Vit D
	RTA	-

Biochemical approach :



mx of CKD associated rickets : ----- Active space -----

1. Control hyperphosphatemia :

- Restrict dietary intake.
- Ca based phosphate binders : Ca acetate.
- Non-Ca based binder : Sevelamer.

2. Vit D Rx :

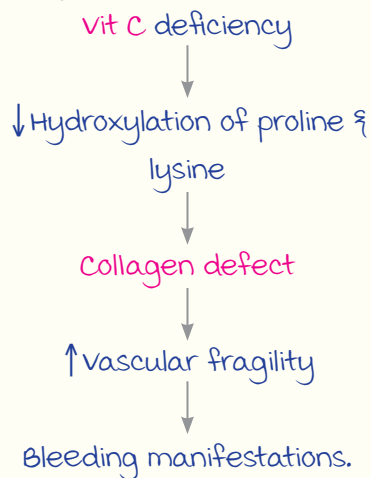
Initial : 25-OH D (Calcitriol)
 No response, PTH ↑
 Active analogues : 1, 25-(OH)₂D₃ (Calcitriol).

Miscellaneous Deficiencies

01:23:14

Scurvy (Vit C Deficiency) :

Pathology :



C/f :

- Gum bleeds.
- Hair :
 - Perifollicular bleeds.
 - Cork - screw hair.
- Bone :
 - Long bone : Subperiosteal bleeding.
 - Pseudoparalysis (D/t pain ++).
- Chest rosary :
 - Costochondral junction bleed.
 - Angulated swelling.
 - Pain ++.

Zinc Deficiency :

C/f  :

- Poor wound healing.
- Diarrhea.
- Growth retardation.
- Hypogonadism.



Gum bleeding

----- Active space ----- Acrodermatitis enteropathica 🐘 :

- Autosomal recessive.
- Defective gene : **SLC 39A4** $\rightarrow$ ZIP 4 (Zn transporter protein) $\rightarrow$ $\downarrow$ Zn uptake.
- c/f :
 - Diarrhea.
 - Rash : **Peri-orificial**.
 - Alopecia.



Perianal rash



Perioral rash



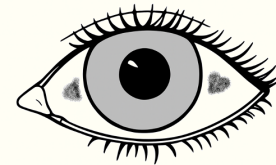
Alopecia

Acrodermatitis enteropathica

Xerophthalmia (Vit A Deficiency) :

c/f :

- Earliest feature : Night blindness ($\downarrow$ Rhodopsin).
- Earliest sign : Conjunctival xerosis.
- Characteristic sign : **Bitot's spots**.
(Silvery, white foamy plaques on bulbar conjunctiva).
- Dreaded complication : **Blindness**.



Bitot's spots

Corneal ulcers $\leftarrow$ Keratomalacia $\leftarrow$ Corneal xerosis

- Leading cause of blindness in pre-school children : Preventable.

Rx :

Dose :

	Dose
< 6 months	50,000 IU
6 - 12 months or < 8 kg	1 lakh IU
> 1 yr	2 lakh IU

- Age of 1 yr + **< 8 kg** : 1 lakh IU.
- Schedule : Day 0, 1, 14.

Prophylaxis :

9 months age to 5 years : Once every 6 months.

CHILDHOOD INFECTIONS

----- Active space -----

Measles

00:00:10

Causative organism : measles virus (RNA virus → Paramyxoviridae).

Age : < 5 yrs (Preschool).

Incubation period : 10 days.

mode of transmission : Droplet.

Secondary attack rate : > 90% (Highly contagious).

Infectivity period : 4 days before to 5 days after onset of rash.

Clinical Features 🐘 :

Day 1 - 3 (AKA Prodrome)	Day 4 - 8
Clinical features : • Fever • Cough • Coryza • Conjunctival congestion • Koplik spots : - Appears on day 2 - 3 - Opposite to 2nd lower molar	maculopapular rash : • Site : Behind ears & face ↓ Trunk and extremities • Disappears after 4 days • Leaves behind brownish pigmentation
 Koplik spots white spot surrounded by reddish hue/erythema (Rice grain appearance)	 maculopapular rash

----- Active space -----

Complications :

Acute :

- m/c : Otitis media.
- most severe : Bronchopneumonia (↑ mortality).

Chronic : Subacute sclerosing panencephalitis (SSPE).

- 7 - 8 yrs after 1° infection ; Rare.
- Features :
 - 1st feature : Deterioration of scholastic performance.
 - myoclonic seizures (EEG : Slow wave periodic complexes AKA Rademecker complexes).
 - Personality disturbances.
- Diagnosis : IgG anti-measles antibodies in CSF.
- Fatal condition.

management :

Vitamin A : ↓ mortality & morbidity.

- Improves immune function.
- Improves integrity of respiratory & gastrointestinal epithelium.
- One dose :

measles vs. Roseola Infantum :

	measles	Roseola infantum/ sixth disease/ exanthema subitum
Fever + day 4 maculopapular rash		
Causative organism	measles virus	HHV-6 > HHV-7
Fever	↑ After onset of rash	↓ After onset of rash
Site of rash	Starts behind ear/face	Starts on trunk (Chest)

Hand, Foot & Mouth Disease (HFMD)

00:09:59

----- Active space -----

Causative organisms : **Coxsackie A16** > Enterovirus 71.

Age : Primary preschoolers.

Clinical features : usually **self resolving**.

- Fever
- Blisters around mouth.
- Papulovesicular blisters on hands & feet.

Treatment : Supportive.



Blisters around mouth



Papulovesicular blisters
in foot & hands

Erythema Infectiosum/5th Disease

00:11:37

Causative organism : Parvovirus B19.

Age group : 5 - 15 yrs (School-going children).

Clinical features :

- Fever.
- Rash :
Redness in the face/cheeks
(**Slapped cheek appearance**)

↓ spread

macular rash with **central clearing** in trunk & limbs
(Lacy/reticulated pattern).



Slapped cheek appearance



Lacy/reticulated pattern

Associations  :

- Arthralgia : In adolescents/older children.
- Transient aplastic crisis : In children with chronic hemolytic anemia.
- maternal infection : **Nonimmune hydrops fetalis**.

----- Active space -----

Varicella/Chicken Pox

00:14:42

Causative agent : Herpes zoster virus.

Age : 5 - 10 yrs (School going children).

Incubation period : 10 - 21 days.

mode of transmission :

Droplet spread/Direct contact with infected lesions.

Infectivity period 🐘 :

24 - 48 hrs before rash till all lesions have crusted.



Fluid filled vesicles

Clinical Features :

Fever + malaise : Prodrome (1 - 2 days).

Rash :

- Starts in trunk → Spreads to other areas.
- macule → Papule → Clear fluid filled vesicles (with surrounding erythema).
(Dew drops in rose petal appearance)
- Pleomorphic appearance.
- Pruritic lesion.

Treatment :

Acyclovir x 5 days : Start within 24 hours.

Congenital varicella :

microcephaly

Limb hypoplasia

Cicatricial scar
(Linear dermatomal
pattern)**Note :**

Linear dermatomal scars : Also seen in shingles.

Diphtheria

00:19:24

----- Active space -----

Causative organism : *Corynebacterium diphtheriae*.

Incubation period : 2 - 5 days.

mode of transmission : Aerosol droplets.

↑ Risk : unimmunized.

Clinical Features :

Appearance : Toxic/sick appearance.

Constitutional symptoms : Fever, sore throat.

Drooling of saliva : ↓ Swallowing d/t pain (Dysphagia).

Neck swelling : d/t cervical lymphadenopathy → Bull neck appearance.

Pseudomembrane : Lacks epithelial structures.

- d/t exotoxin → Exudative necrosis + fibrin deposition.
- Occurs in throat → Can extend to tonsils, uvula, posterior pharynx.
- Bleeds on removal.

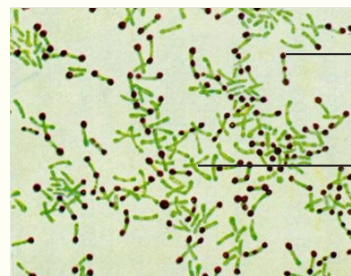


Pseudomembrane

management :

Diagnosis : Oropharyngeal swab

↓
Albert stain.



Albert stain

metachromatic granules

Chinese letter pattern (Bacilli arranged at angles with each other)

Treatment :

- Diphtheria antitoxin (Prevents systemic complications).
- Antibiotics :
 - macrolides (Azithromycin/Erythromycin).
 - Penicillins (Ampicillin).

management of Contacts :

monitoring & isolation : For 7 days.

Chemoprophylaxis :

One dose benzathine penicillin (Im) OR Oral erythromycin x 7 days.

Diphtheria toxoid :

- Indications :
 - unimmunized contacts.
 - < 3 doses of toxoid.
- Regimen
 - < 7 years → DPT.
 - > 7 years → Tdap.

----- Active space -----

Pertussis/Whooping Cough

00:27:27

Causative organism : *Bordetella pertussis*.

Age : Infants & young children.

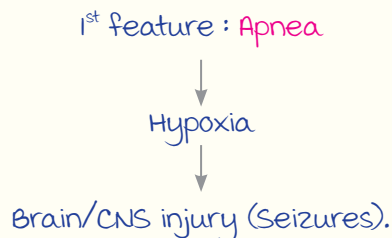
Secondary attack rate : $\approx 100\%$.

mode of spread : Aerosol droplets.

Clinical Features :

Catarrhal stage	Paroxysmal stage	Convalescent stage
• 1 - 2 weeks • Highly infectious • C/f similar to URTI : Fever, running nose, cough (Cough $\uparrow$ intensity as disease progresses)	• 2 - 6 weeks • uncontrollable episodes of cough f/b whoop (Sudden gush of air through partially closed glottis)	Recovery (> 6 weeks)

Infants :

**management **

Investigations :

- Blood count : Lymphocytic leukocytosis ($> 50,000/\text{mm}^3$).
- PCR.
- Culture :
 - Oropharyngeal swab.
 - Cough plate culture : Bordet Gengou media.

Treatment :

- macrolide antibiotics (Azithromycin, Clarithromycin, Erythromycin).
- Erythromycin avoided in neonates (A/w pyloric stenosis).

COVID-19 in Children

00:33:01

----- Active space -----

Severity	Symptoms	Treatment
mild/ asymptomatic (m/c)	mild constitutional symptoms	Symptomatic treatment
Severe	Low SpO_2 + Any one of the following : • ARDS • MODS • Severe pneumonia • Shock • Seizures	• Steroids (Dexamethasone/ methylprednisolone) - Given for 5 - 7 days - Tapered over 10 - 14 days • No role of antivirals

TORCH Infections

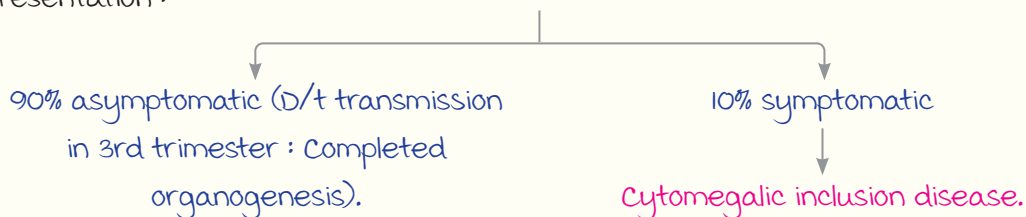
00:35:03

- Transmission of infection during gestation.
- ↑R/o congenital anomalies & **brain defects** (Low IQ, seizures).
- T : Toxoplasmosis.
- O : Others (Syphilis, Zika, Parvovirus-B19).
- R : Rubella.
- C : CMV (m/c).
- H : Herpes zoster.

Congenital CMV

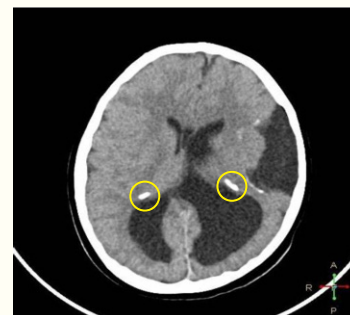
↑Risk : **Primary** infection of mother with CMV during pregnancy.

Presentation :



Clinical features :

- Low IQ, microcephaly (D/t brain defects).
- Chorioretinitis.
- Petechiae (D/t thrombocytopenia).
- Hepatosplenomegaly.



Periventricular calcification

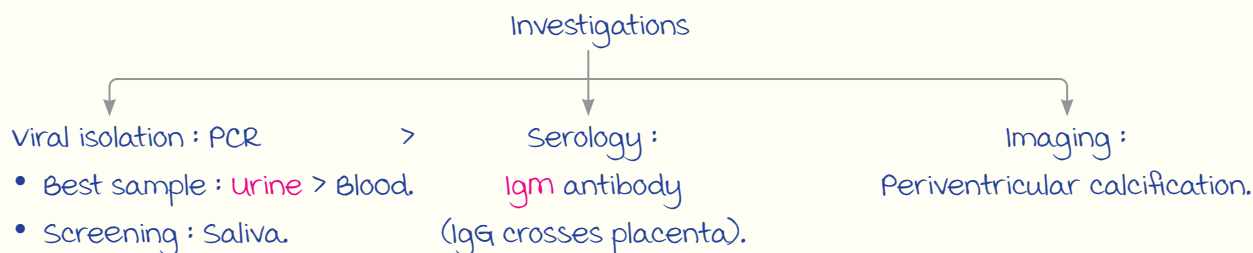
----- Active space -----

Long term sequelae : SNHL.

- m/c cause of non-syndromic hearing loss.

Investigations :

Within 2 - 3 weeks after birth → To differentiate b/w congenital & post natal CMV infection.

Congenital Toxoplasmosis :Clinical features  :

- SNHL
 - Chorioretinitis (m/c)
 - microcephaly
 - Hydrocephalus (Few infants).
 - Diffuse parenchymal calcification.
- Triad of congenital toxo.
- } Similar to CMV.



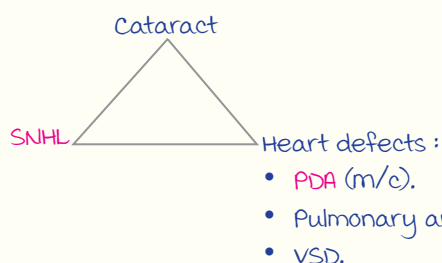
Diffuse parenchymal calcification

Congenital Rubella :

most severe TORCH infection.

Clinical features :

- Triad :

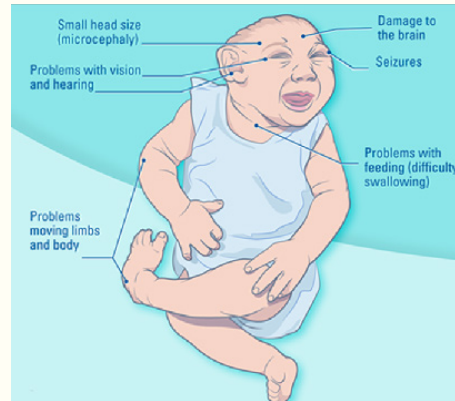


- Salt & pepper retinopathy (m/c eye finding).

Congenital Zika Infection :

Clinical features :

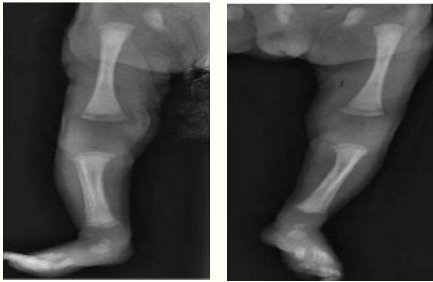
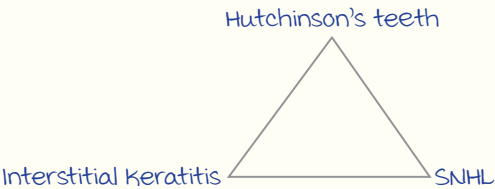
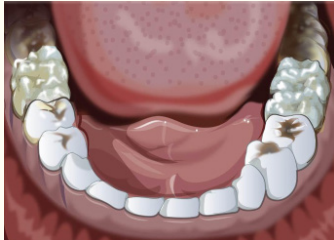
- Limb defects : D/t contractures.
- microcephaly.
- **Subcortical calcification.**



Congenital zika infection

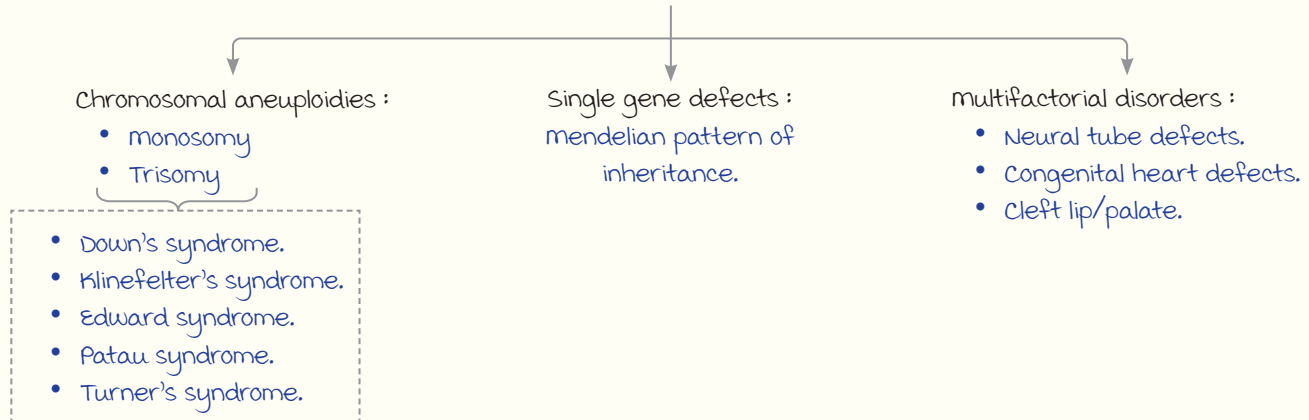
----- Active space -----

Congenital Syphilis :

Early congenital syphilis	Late congenital syphilis 
• Vesiculobullous lesions in soles/palms • Snuffles : Rhinitis with ↑↑ discharge • Pseudoparalysis : Inflammation of bone → ↑ Pain  Vesiculobullous lesions in sole  Pseudoparalysis	Hutchinson's triad : Hutchinson's teeth  Interstitial keratitis SNHL  Hutchinson's teeth (Notching/peg shaped incisors)  mulberry molars (Rough appearance)

GENERAL PAEDIATRICS : GENETICS

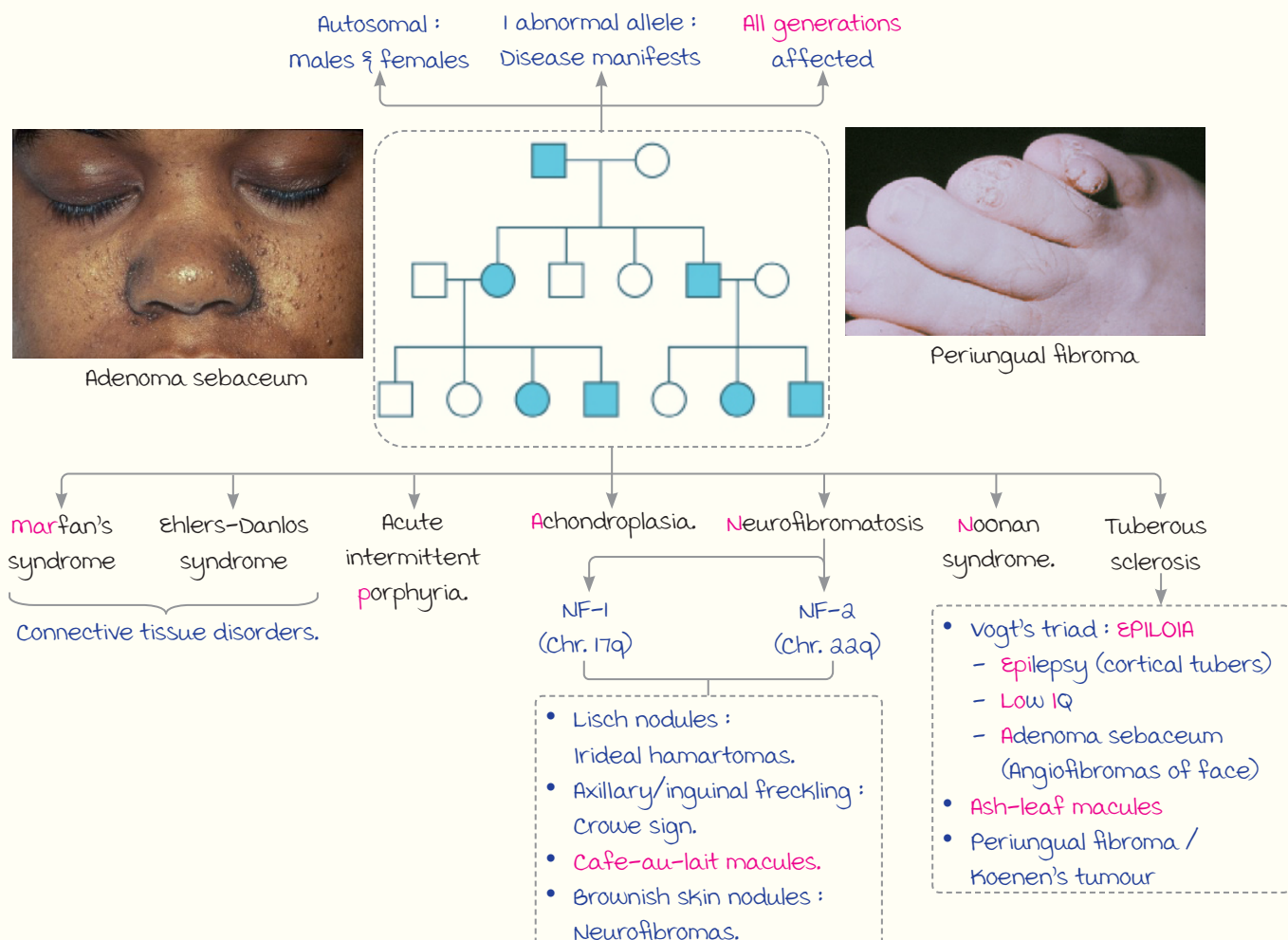
Classification :



Mendelian Modes of Inheritance

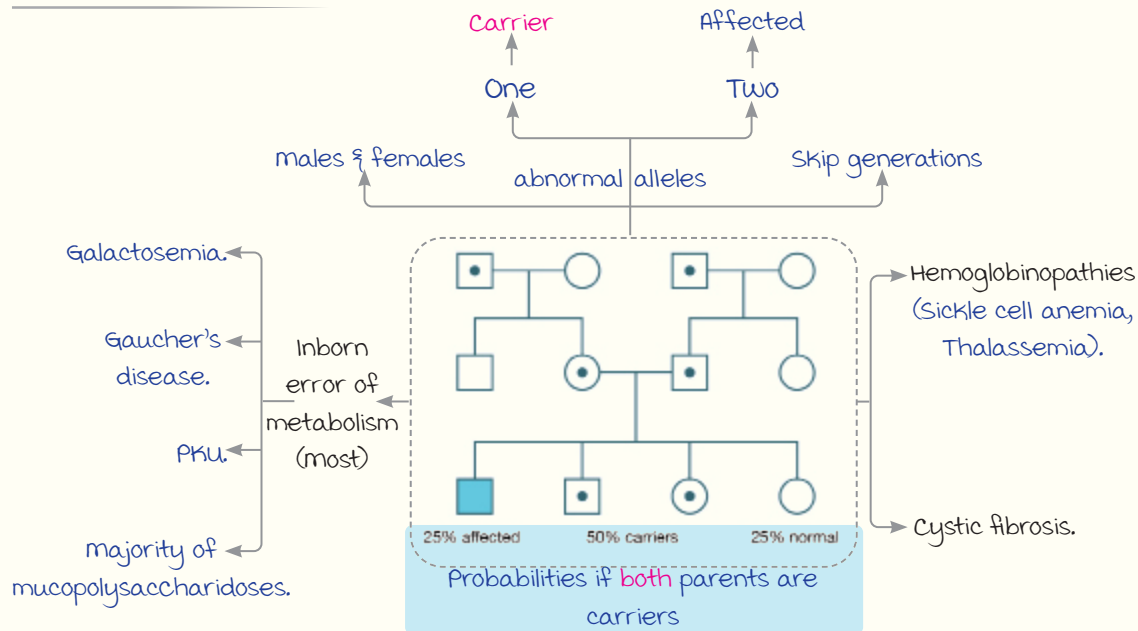
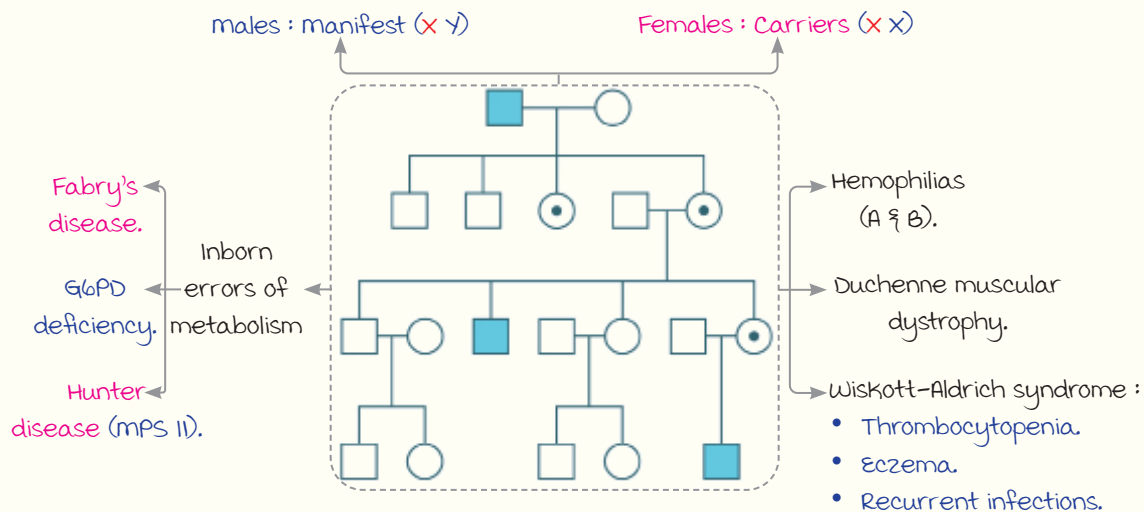
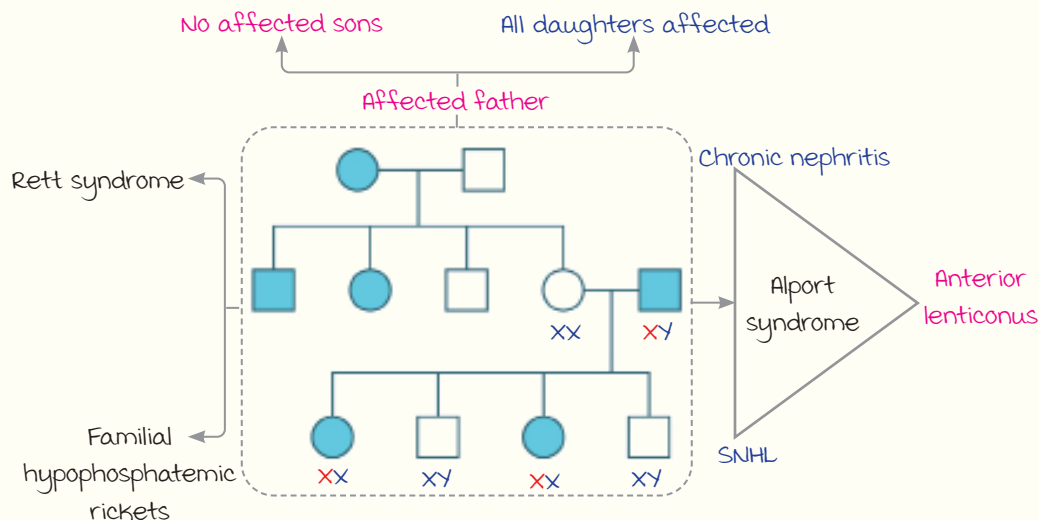
00:01:39

Autosomal Dominant :



Autosomal Recessive :

----- Active space -----

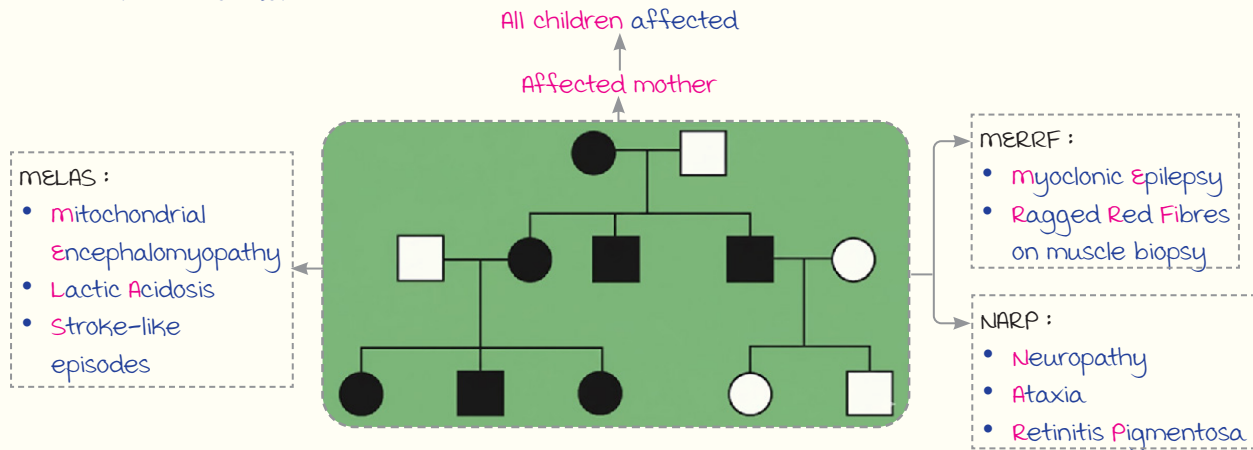
X-linked Recessive :X - linked Dominant :

----- Active space -----

Mitochondrial Inheritance

00:22:26

Non-mendelian.

**Down Syndrome**

00:27:07

m/c genetic disorder.

Genetic Defect :

Trisomy 21 (95%) :

↓/↑ maternal meiotic non-disjunction

(↑Risk : mother > 35 yrs).

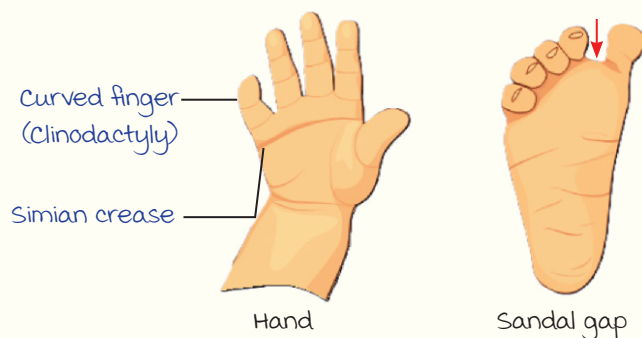
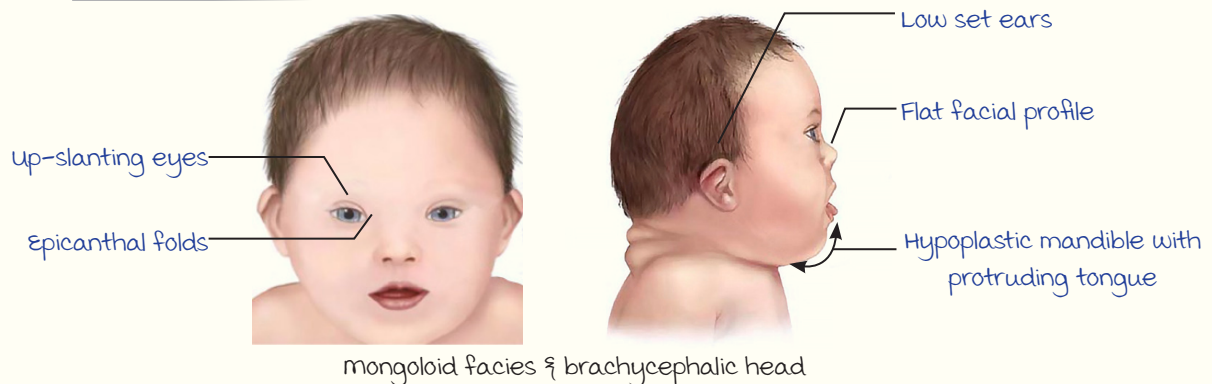
Robertsonian translocation (4%)  :

- t (14 ; 21).
- t (21 ; 21) :

100% Down syndrome in offspring.

mosaic (1%) :

milder features.

External Features :

Brushfield's spots :

Condensed connective tissue in iris

Associated Comorbidities :

----- Active space -----

- Heart defects :
 - Seen in 40% patients.
 - 40 - 60% : Endocardial cushion defects (ASD + VSD), valve defects.
- Brain :
 - Low IQ.
 - Hypotonia.
 - microcephaly.
- GI defects :
 - Atresia : Duodenal atresia (m/c) → X-ray abdomen : Double-bubble sign.
- Premature Alzheimer's disease.
- ↑ Risk of leukemia : ALL > AML.
 - Exception : < 3 yrs → AML-M7 (m/c).

Antenatal Diagnosis :

Best confirmatory test : Karyotyping.

- Sample used :
 - Chorionic villus sampling (CVS)
 - Amniocentesis
 } Risk of abortion.

Screening :

- USG : End of 1st trimester.
 - Nuchal thickness (NT) : > 3 mm.
 - Absent/hypoplastic nasal bone.
- markers :
 - T₁ : PAPP-A + β-hCG (Double test).
 - T₂ : β-hCG + AFP + unconjugated estriol (Triple test).
 - T₃ : Triple test + Inhibin A (Quadruple test).
- Best screening method : Integrated test (95% detection).
Age + NT/PAPP-A (T₁) + Quadruple test (T₃)

PAPP-A : Pregnancy associated plasma protein-A.

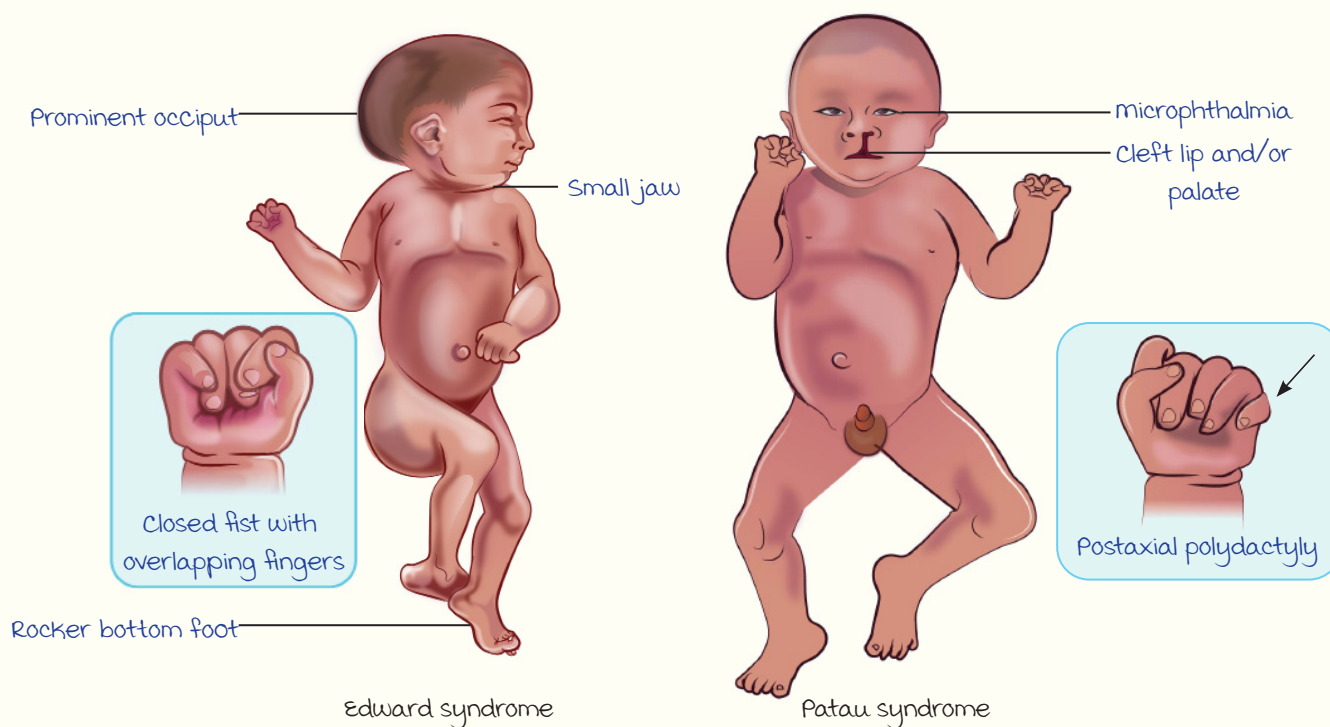
AFP : Alpha fetoprotein.

----- Active space -----

Edward Syndrome & Patau Syndrome

00:41:10

	Edward syndrome (2 nd m/c trisomy)	Patau syndrome
Trisomy	18	13
Common features	• Low IQ • microcephaly • Heart defect : VSD (ventricular septal defect)	
Characteristic features	• Prominent occiput • micrognathia • Closed fist with overlapping fingers • Rocker bottom foot	• microphthalmia • Aplasia cutis (skin absent on parts of scalp) • Postaxial polydactyly • Holoprosencephaly (undivided forebrain) • midline facial defects : - Cleft lip/palate - Proboscis - Cyclopia (fused eye)



Klinefelter Syndrome

00:46:57

----- Active space -----

Chromosomal Abnormality :

47, XXY (Additional X chromosome).

Salient Features :

Low IQ.

Disproportionate tall stature :

D/t 2 copies of SHOX gene

(Stature homeobox gene) on X chromosomes.

micro-orchidism :

↓ Testosterone → Delayed puberty.
→ Infertility.

Gynaecomastia :

↑ Aromatase → Relative ↑ in estrogen metabolites.



Klinefelter syndrome

Turner Syndrome

00:50:20

AKA monosomy X : 45 XO.

Salient Features :

- Normal IQ.
- Short stature.

Early features :

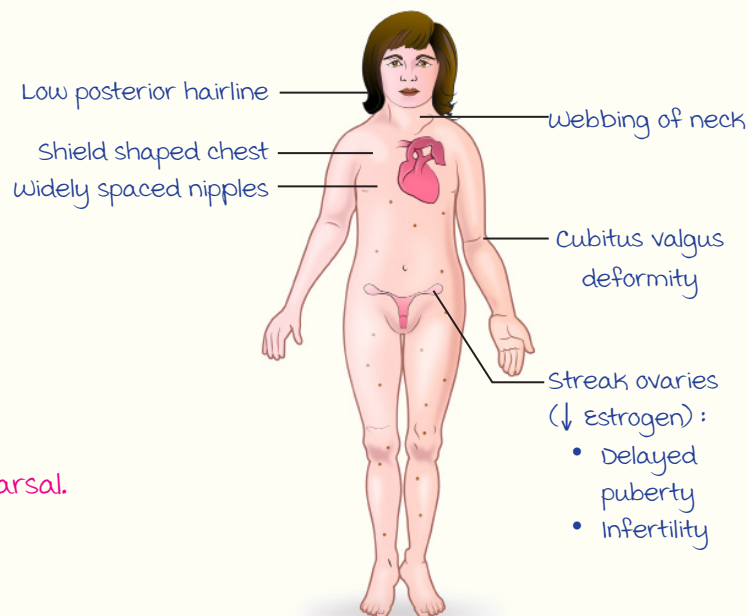
- Lymphedema.
- Limbs.
- Neck (Cystic hygroma).

X-ray :

Short 4th metacarpal/metatarsal.

Heart defect :

- Bicuspid aortic valve.
- Coarctation of aorta.



Complete manifestation : mid-childhood

----- Active space -----

DiGeorge Syndrome

00:54:50

Genetic Defect :

microdeletion on Chr. 22q



Defective development of 3rd & 4th pharyngeal pouches.

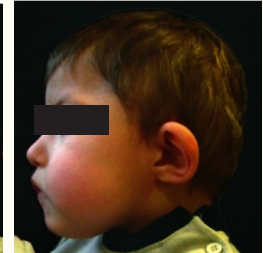
Salient Features 🐘 :

mnemonic : **CATCH 22**

- Cardiac defects (Conotruncal defects) : Tetralogy of Fallot.
- Abnormal facies.
- Thymic aplasia (T-cell immunodeficiency).
- Cleft lip/palate.
- Hypocalcemia (↓ PTH).
- 22 - chromosome affected.



Thin upper lip



Low set ears

METABOLIC DISORDERS

----- Active space -----

Carbohydrate Metabolism Related Disorders

00:00:15

Galactosemia :

- m/c carbohydrate related metabolic disorder.
- Autosomal recessive.

Enzyme deficient :

Galactose - 1 - PO_4 uridyl transferase (GALPUT) $\rightarrow$ Accumulation of galactose - 1 - PO_4 .

Clinical features  :

Liver	Eye	Kidney
• Chronic liver disease • Hepatomegaly	Galactose $\xrightarrow{\text{Aldose reductase}}$ Galactitol $\downarrow$ Oil drop cataract	Proximal tubule dysfunction

Investigations :

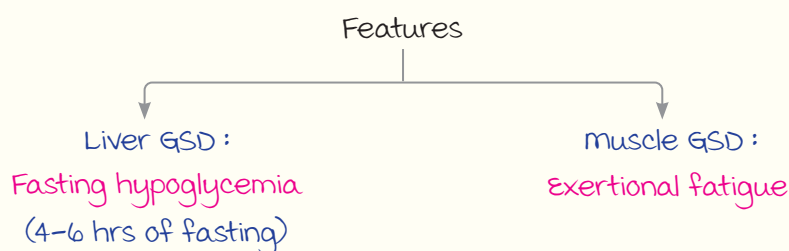
- urine : Galactose (Reducing sugar) $\rightarrow$
 - Benedict's test +ve.
 - Glucose oxidase test -ve.
- $\downarrow$ GALPUT enzyme : Confirmatory test.

Complication  : Prone to *E. coli* sepsis (Fatal).

management :

- Breastfeeding is c/i.
- Lactose free formula feeds.

Glycogen storage disorder (GSD) :



----- Active space ----- Types  :

Liver GSD	
Type	Enzyme deficient/defect
0	Glycogen synthase
I : von Gierke's disease (m/c)	Glucose - 6 - Phosphatase
3 : Cori's disease	Debranching enzyme
4 : Andersen's disease	Branching enzyme
6 : Her's disease	Hepatic phosphorylase
II : Fanconi - Bickel syndrome	GLUT - 2

muscle GSD	
Type	Enzyme deficient/defect
2 : Pompe's disease	α -glucosidase
5 : mc Ardles disease (m/c)	muscle phosphorylase
7 : Tarui disease	Phosphofructokinase

Von Gierke's disease :

- Defect : Glycogen $\rightarrow$ Glu-6-PO₄ $\xrightarrow[\text{PO}_4]{\text{Glucose-6-phosphatase}}$ $\downarrow$ Glucose.
- C/F :
 - Fasting hypoglycemia.
 - Hyperlipidemia $\rightarrow$ Doll like facies.
- management :
 - Frequent feeds x 3-4 hours.
 - Overnight nasogastric tube feeding.
 - Uncooked corn starch (Before bed) : Slow release of glucose.



Doll - like facies

Pompe's disease  :

- muscles affected :
 - Skeletal $\rightarrow$ Exertional fatigue
 - Cardiac $\rightarrow$ Cardiomyopathy $\rightarrow$ Cardiac failure
- ECG :
 - Short P-R interval
 - Large QRS.
- Other C/F :
 - Hypoglycemia.
 - macroglossia.
- management : Enzyme replacement therapy (ERT).

Amino Acid Metabolism Disorders

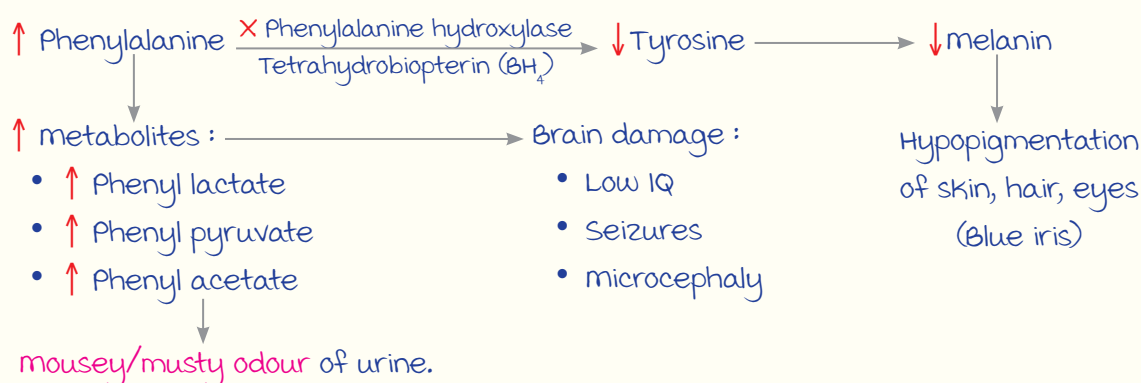
00:14:36

----- Active space -----

Phenylketonuria :

- m/c amino acid disorder.
- Autosomal recessive.

Pathophysiology & clinical features :



Investigations :

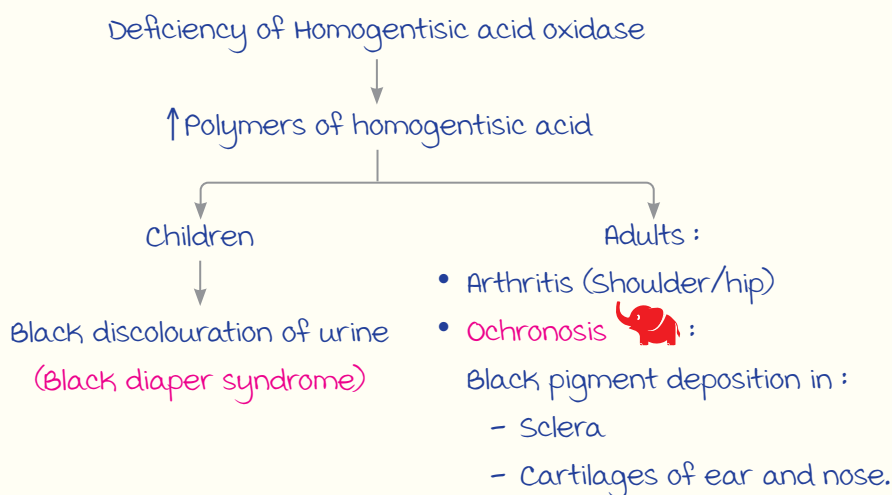
- ↑ Phenylalanine.
- Phenylalanine : Tyrosine → > 3.

management :

- Restriction of phenylalanine.
- Trial of sapropterin (Synthetic BH₄).

Alkaptonuria :

Pathophysiology & clinical features :



Lysosomal Storage Disorder (LSD)

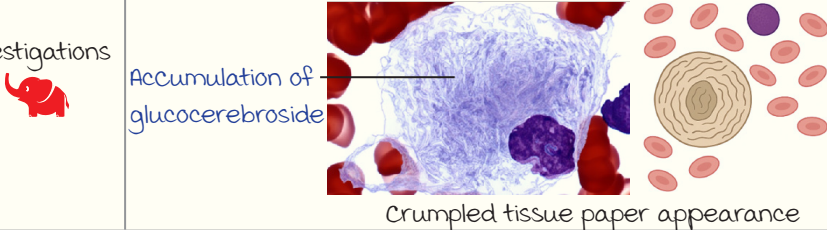
00:21:02

Gaucher's Disease :

- m/c LSD.
- Autosomal recessive.

Deficient enzyme : Glucocerebrosidase.

----- Active space ----- Classification :

	Non-neuronopathic (m/c) :	Neuronopathic
Type	Type I	Type II, III
Clinical feature	- Short stature - Hepatosplenomegaly - Lytic lesion of bone → Bone pain	- Developmental delay - Nystagmus - Hypertonia
Investigations	- X-ray : Erlenmeyer flask deformity - Biopsy : Gaucher cell  Accumulation of glucocerebroside Crumpled tissue paper appearance	-
management	ERT	

Disorders of Copper Metabolism

00:25:31

Wilson's Disease :

Autosomal recessive

Defect : ATP 7B gene on chromosome 13 (Long arm)

↓ ↓ Ceruloplasmin

↑ ↑ unbound free copper → Toxic effects

Clinical features :

Liver	Neurological	Eyes
- Seen in 1st decade - Jaundice - Chronic liver disease → Hepatic failure	- m/c in adults - Wing beating tremors - Dysarthria - Psychosis/depression - Personality disorders	- K-F ring : Cu^{2+} deposition in descemet's membrane of cornea - Sunflower cataract

Investigations :

1. Screening :

- S. Ceruloplasmin ↓
- urine copper ↑

2. Diagnostic :

Liver biopsy : > 250 μg copper/g dry weight of liver.



KF ring

management :

- Copper chelating agents :
 - D-penicillamine (Children).
 - Trientine.
- maintenance : Zinc (↓ Intestinal absorption of copper).

----- Active space -----

menke's Disease :

X-linked recessive

Pathogenesis :

Defect in **ATP 7A gene** → ↓ Activity of copper transporting ATPase → Copper deficiency.

Clinical features :

1. Neurological :

- Low IQ.
- Seizures.
- microcephaly.

2. Hair :

- Brittle hair (menke's kinky hair disease).
- **Pili torti.**
- microscopy :
 - **Fracture of hair shaft.**
 - **Trichorrhexis nodosa.**

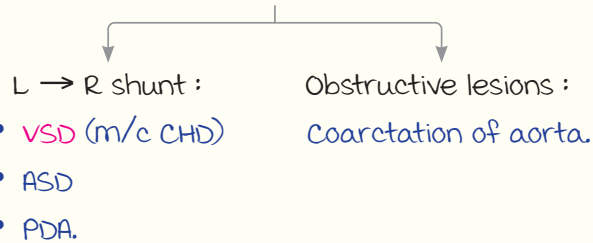
Acyanotic CHD

00:05:00

----- Active space -----

m/c type of CHD

Classification :



ASD : Atrial Septal Defect
 VSD : Ventricular Septal Defect
 PDA : Patent Ductus Arteriosus
 CHD : Congenital Heart Disease
 HF : Heart Failure

L - R shunt :

	m/c type	Characteristics
VSD (m/c)	Perimembranous VSD	• Pansystolic murmur • Site : Left 4th ICS, parasternal area
ASD	Ostium secundum ASD	• Wide, fixed split S_a (D/t late P_a) • No murmur
PDA	-	• Continuous, loud, machinery murmur • Site : Left 2nd ICS

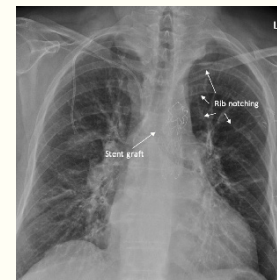
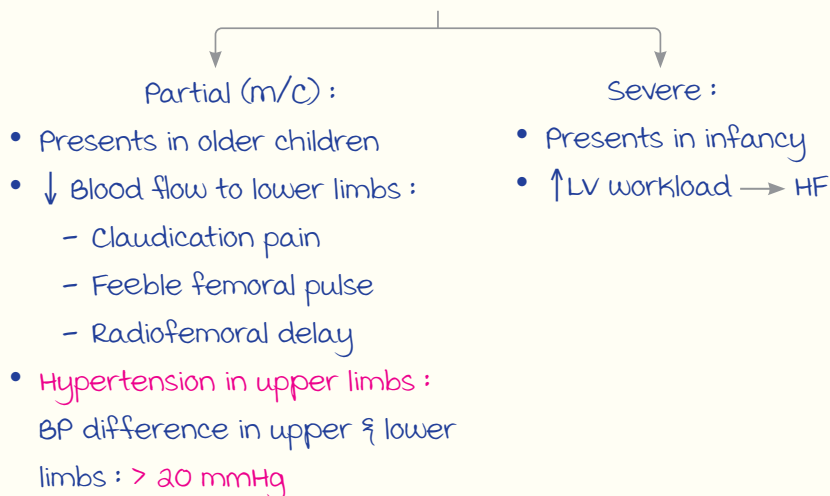
management of PDA :

In preterm babies, medical mx can be tried :

- NSAIDs (DOC) d/t COX $\ominus$ → ↓ Prostaglandin.
- Ibuprofen >> Indomethacin (As Ibuprofen has ↓ risk of nephrotoxicity).

Coarctation of aorta :

- Partial narrowing of aorta → ↓ Flow in descending aorta.
- Type : Juxtaductal (m/c).
- Features :



Inferior rib notching.

----- Active space -----

• Compensation :

Collaterals formation $\longrightarrow$ $\uparrow$ size intercostal arteries $\longrightarrow$ Inferior rib notching
(Seen in > 10 yrs of age)

Cyanotic CHD

00:20:51

Classification :

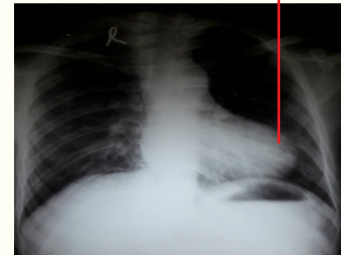
$\uparrow$ Pulmonary blood flow	$\downarrow$ Pulmonary blood flow (m/c)
- Transposition of great arteries (TGA) : m/c cyanotic CHD in neonates - Truncus arteriosus (Single artery from heart) - Total anomalous pulmonary venous communication (TAPVC)	- Tetralogy of Fallot (TOF) : m/c cyanotic CHD overall - Ebstein anomaly - Tricuspid atresia

Tetralogy of Fallot 🐘 :

Components :

1. Overriding of aorta.
2. Right ventricular hypertrophy (RVH).
3. **Infundibular pulmonary stenosis**
(m/c cause of cyanosis in TOF)
4. VSD

upturned apex due to RVH



Boot shaped heart on X ray
(Coeur en sabot)

Complications :

- Polycythemia (D/t chronic hypoxia).
- Cyanotic spells : Infundibular spasm $\xrightarrow{\text{worsens}}$ Cyanosis.

management :

- **β blockers** : $\downarrow$ Infundibular spasms.
- O_2 supplementation.
- $NaHCO_3$: For acidosis
- morphine : $\downarrow$ Hyperventilation.
- α agonists
- Squatting/knee chest posture
- Surgical management
 - $\longrightarrow$ Definitive
 - $\longrightarrow$ Shunts :
 - To $\uparrow$ pulmonary blood flow.
 - **modified Blalock Taussig shunt (m/c)** :
Shunt b/w Subclavian & pulmonary artery.

Tricuspid Atresia :

Pathophysiology :

Atresia of tricuspid valve $\longrightarrow$ $\downarrow$ Blood flow to RV $\longrightarrow$ $\downarrow$ Blood flow to PA $\longrightarrow$ Cyanosis

$\downarrow$
 $\uparrow$ Blood in RA $\xrightarrow[\text{(or) ASD}]{\text{Patent foramen ovale}}$ Blood enters LA & LV $\longrightarrow$ Hypertrophy of left heart.

ECG Findings :

- Left Axis deviation.
- Evidence of LVH.

Ebstein's Anomaly  :

A/w : maternal **lithium** exposure.

Pathophysiology :

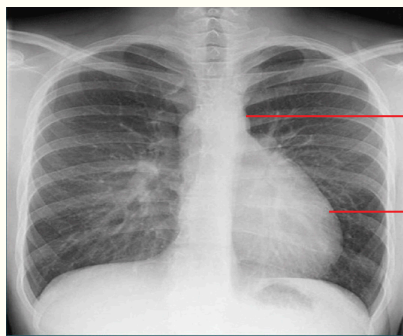
Inferiorly displaced → Atrialization of RV → ↑ size of RA → Box shaped heart.
tricuspid valve



Box shaped heart

----- Active space -----

Transposition of Great Arteries (TGA) :



Egg on string appearance

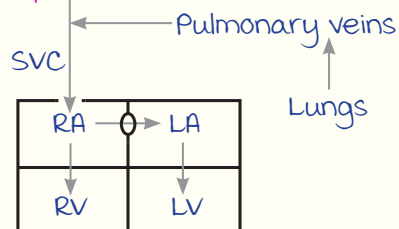
Interchanged position of Aorta & pulmonary artery

Narrow vascular Pedicle (String)

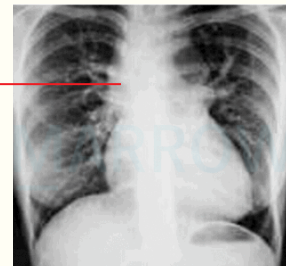
Heart shadow (Egg)

Supracardiac TAPVC :

O_a saturation equal in all 4 heart chambers.



Dilated SVC



Snowman/Figure of 8 appearance

Acute Rheumatic Fever

00:42:10

Etiology :

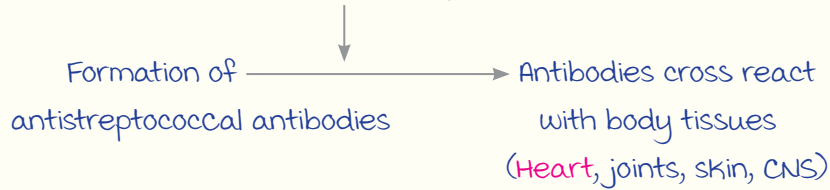
Age : 5 - 15 years.

Post streptococcal infection :

- Group A β -hemolytic streptococcus.
- Pharyngitis After 2-3 weeks → Rheumatic fever.

----- Active space -----

Pathogenesis : molecular mimicry

modified Jones Criteria (2015) :

For high risk population :

major	minor
• Pancarditis (90%) : Clinical/subclinical • Arthritis (mono/poly) or polyarthralgia (30 - 50%) • Erythema marginatum (Early, non pruritic) • Subcutaneous nodules (Late, non tender) • Sydenham's chorea (Last feature)	• monoarthralgia • Fever $\geq 38^{\circ}\text{C}$ • ESR $\geq 30 \text{ mm/h}$ in 1st hr OR CRP $\geq 3.0 \text{ mg/dL}$ • Prolonged PR interval

Dx :

1. 2 major OR 1 major + 2 minor OR 3 minor (In recurrence)

+

2. ↑ ASO titre

Treatment :

Bed rest.

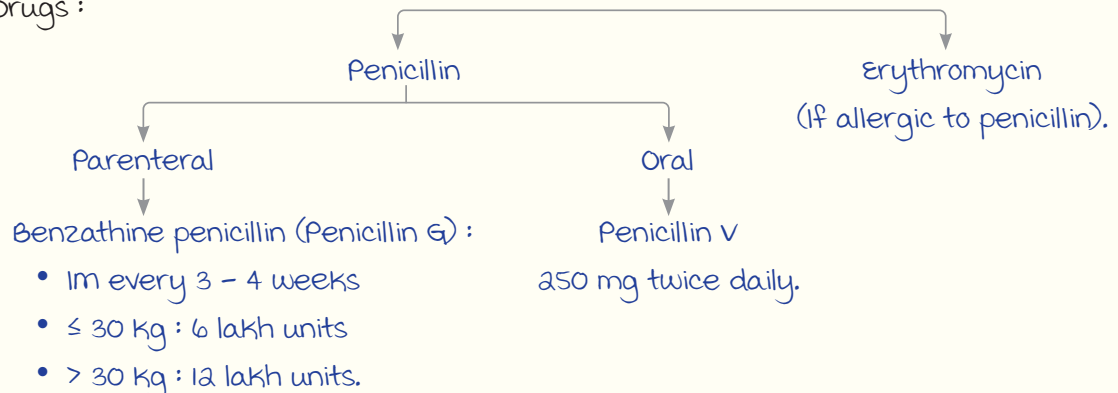
Antibiotics : Inj. Benzathine Penicillin (Im) : Single dose
(or)

Oral Penicillin V : 10 days

Anti-inflammatory drugs : Steroids > Aspirin
(Duration : 12 weeks)Prophylaxis :

2° prevention.

Drugs :



Duration  :

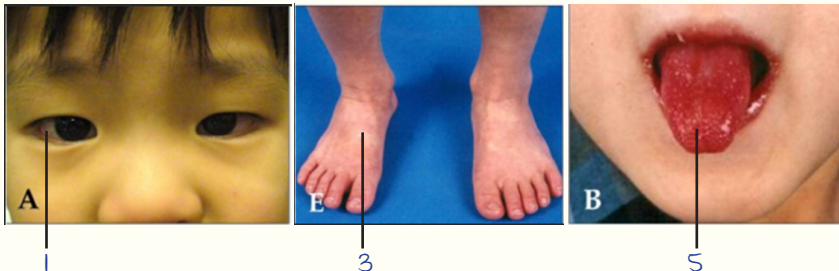
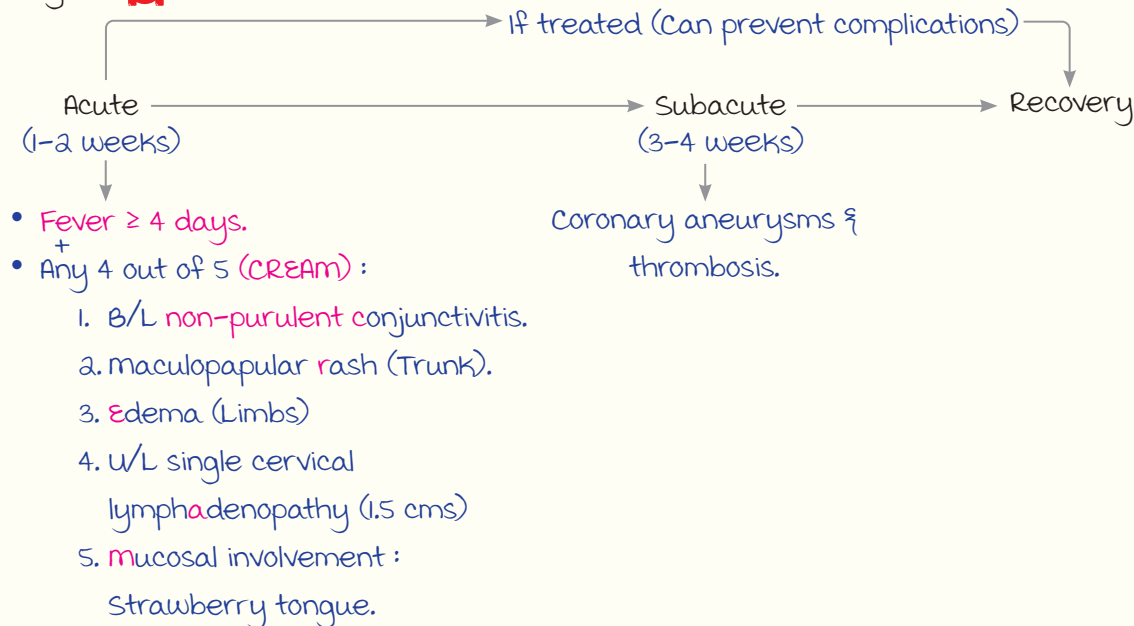
----- Active space -----

	Duration (whichever is longer)
Without carditis	5 yrs or till 21 yrs
With carditis	10 yrs or till 21 yrs
With residual heart disease	10 yrs or till 40 yrs/ Lifelong

Kawasaki Disease

00:53:27

- Age < 5 years.
- medium vessel vasculitis : Affects coronary artery (20 - 30%).
- **m/c childhood vasculitis in India.**

Stages  :**Treatment :**IVIg : **DOC.**

Aspirin :

- 30-50 mg/kg/day (Till pt is afebrile for 48-72 hrs)
- ↓
- Low dose : 3-5 mg/kg/day (for 6-8 weeks).
- Lifetime low dose : Rx of coronary artery aneurysms.

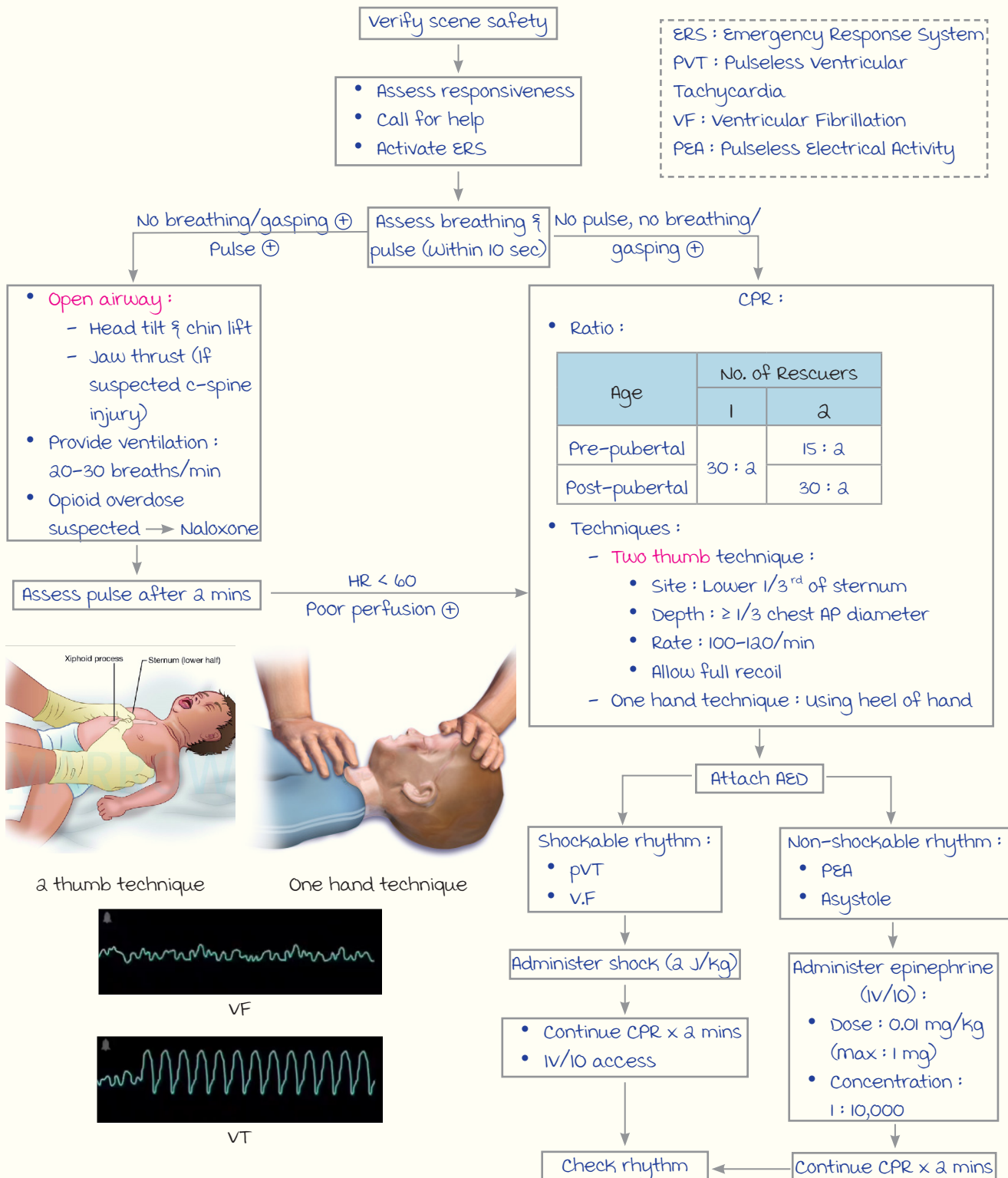
----- Active space -----

Intensified therapy :

- used for high risk cases
 - Infant < 6 months
 - Abnormal echo
- **IVIg + Steroids/Infliximab.**

Pediatric Resuscitation

01:00:39



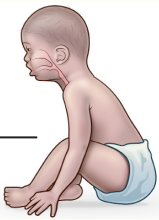
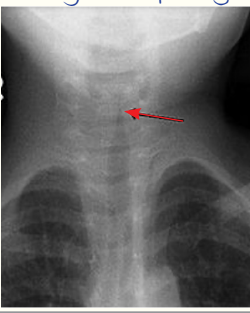
SYSTEMIC PAEDIATRICS : PULMONOLOGY

----- Active space -----

Respiratory Infections

00:00:07

Upper Airway Infections 🐘:

	Acute epiglottitis	Croup : ALTB (A/c laryngotracheobronchitis)
m/c etiology	<i>H. influenzae type B</i>	- Parainfluenza virus - Influenza virus - Adenovirus
Presentation	- Toxic appearance - Sore throat, dysphagia - Drooling of saliva ⊕ - Tripod position - Stridor - Fever 	- Well appearance - Low grade fever - Slow progression
Investigations	- Laryngoscopy : Cherry red appearance of epiglottis - X-ray : Thumbprint sign 	X-ray : Steeple sign 
Rx	- Secure airway - IV ceftriaxone	- Dexamethasone (Single dose) - Racemic epinephrine nebulisation : moderate-severe croup (Retractions ⊕, ↓SpO₂)

Lower Airway infections :

	Acute bronchiolitis	Pneumonia
m/c etiology	Respiratory syncytial virus (RSV)	Bacteria : <i>Streptococcus pneumoniae</i> > Virus : RSV (< 2 yrs)
Presentation	- Fever - Breathing difficulty	
	Wheeze ⊕	Crepitations ⊕

----- Active space ----- Pneumonia : IMNCI guidelines 

	Features	management
No pneumonia	Fever, cough/cold	- At home : Paracetamol (PCT) - Follow up : After 5 days
Pneumonia	- Fever, cough/cold + fast breathing and/or chest indrawing - SpO₂ : Normal	- At home : PCT + Oral amoxicillin x 5 days - Follow up : After 2 days
Severe pneumonia AKA very severe disease	- SpO₂ < 90% OR - Stridor in calm child OR ≥ 1 danger signs : - Lethargy - Not feeding - Convulsions - Vomiting	urgent referral to tertiary care hospital + 1st dose : IV antibiotic (Ampicillin + gentamicin)

Fast breathing :

Age	RR (Per min)
< 2 months	> 60
2 – 12 months	> 50
> 12 months	> 40

Foreign Bodies in Airway

00:12:50

Paediatric emergency.

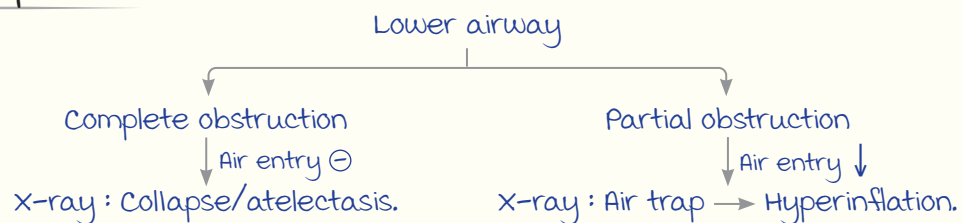
Age group: usually < 5 years.

Location: Lower airway > Upper airway

Trachea, bronchus (R > L) Larynx

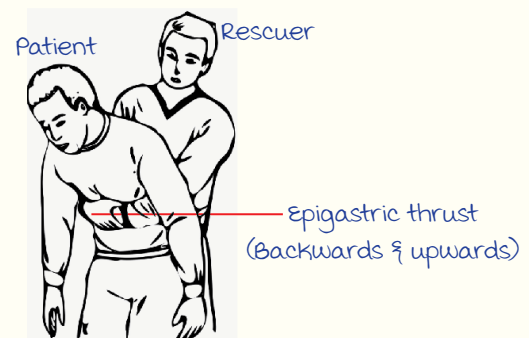
Presentation : Sudden breathing difficulty.

Consequences :



management  :

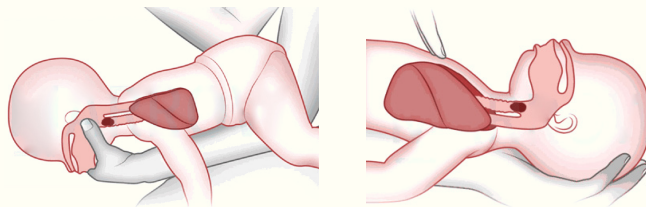
1. Lower airway foreign body (Bronchus) → Rigid bronchoscope removal.
2. upper airway foreign body (Larynx):
 - > 1 year : 5 back blows + 5 epigastric thrusts (Heimlich's maneuver).



Heimlich's maneuver

- < 1 year : 5 back blows + 5 chest thrusts (Alternate) → unresponsive → CPR.

----- Active space -----



Asthma

00:19:56

- Chronic inflammatory disorder of airways.
- Hyper-responsiveness of airways → Bronchoconstriction.

Trigger :

- Allergen.
- Infections.

Association :

- Atopic dermatitis.
- Allergic rhinitis.

Investigations :

- $\downarrow$ FEV₁/FVC ratio : < 0.8 (Obstructive)
- Response to inhaled β_a agonist :
 $> 9 - 12\% \uparrow$ in FEV₁.
- Diurnal variation in PEF or FEV₁ : $\geq 20\% \uparrow$.
- Exercise challenge : $\geq 15\% \downarrow$ in FEV₁.

Symptoms :

- Wheeze.
- Cough.
- Chest tightness.
- Dyspnea.

management :

	Symptoms	Treatment
Infrequent asthma	≤ 2 episodes/week	Relievers (SOS) : SABA + ICS (Budesonide)
Frequent asthma	≥ 2 episodes/week	Low dose ICS (Daily)
	Daily episodes (Not continuous)	medium dose ICS (Daily) $\downarrow$ Not controlled LABA (Daily) : • Salmeterol • Formoterol

ICS : Inhaled corticosteroids ; SABA : Short acting β_a agonist ;
LABA : Long acting β_a agonist.

Biological agents : use → Refractory asthma.

Agents	MOA
Omalizumab	Anti-IgE
mepolizumab	Anti-IL-5
Benralizumab	Anti-IL-5 $\mathbb{R}$ - α
Dupilumab	Anti-IL-4 $\mathbb{R}$ - α

----- Active space -----

management of acute exacerbation :

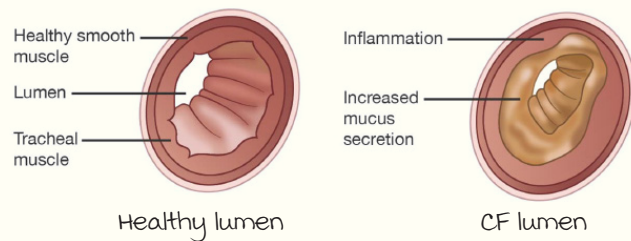
- First line :
 - a. O_2 Supplementation : $SpO_2 \geq 94\%$.
 - b. Salbutamol nebulization : Once every 20 mins in 1st hr (3 times).
 - c. Systemic steroid : Oral prednisolone or IV hydrocortisone.
- Second line :
 - a. $MgSO_4$.
 - b. Terbutaline.
 - c. Theophylline.

Cystic Fibrosis (CF)

00:29:10

- Autosomal recessive.
- CFTR mutation (Cystic fibrosis transmembrane regulator) : Chr. 7q.
- Del F508 mutation (m/c) : Deletion of phenylalanine at 508 position.
- AKA mucoviscidosis.

Defect :

 Cl^- channel inactivation.

Clinical features :

	manifestations
RS	• Recurrent pneumonia (m/c) : - Staph. aureus (m/c) - Pseudomonas aeruginosa • Bronchiectasis • Nasal polyp
GIT 	• meconium ileus (early manifestation) : Soap bubble/ground glass appearance (X-ray) • Distal intestinal obstruction syndrome (DIOS) : Older children • malabsorption • Pancreatic insufficiency (In 85% of cases) - Steatorrhea - ↓ vit A, D, E, K absorption } ↓ Fat absorption (exocrine) - Diabetes (endocrine) • Neonatal cholestasis, biliary cirrhosis
Miscellaneous	• B/L absent vas deferens → Azoospermia → male infertility • Excessively salty sweat (D/t ↑ Na, ↑ Cl) → Contraction alkalosis

Diagnostic criteria :

----- Active space -----

A. Suspect if :

Typical clinical features (RS, GIT or GU) OR Family history OR Immunoreactive trypsinogen test (IRT) in newborn screening ⊕

B. Confirmatory :

- a elevated sweat chloride on separate days $> 60 \text{ mEq/L}$ OR
- a CFTR mutations ⊕ : Best investigation.

Treatment :

- CFTR modulators :

CFTR potentiator :

- Improve Cl^- channel conduction.
- E.g. : Ivacaftor.

CFTR corrector :

- modifies protein folding.
- E.g. :
 - Lumacaftor.
 - Tezacaftor.
 - Elexacaftor.

- Approved Rx : Ivacaftor + Tezacaftor + Elexacaftor.

SYSTEMIC PAEDIATRICS : GASTROENTEROLOGY

Esophageal Disorders

00:00:10

Esophageal Atresia :

Association :

Tracheo-esophageal fistula (TEF) : Air entry → Trachea Fistula → Esophagus → Stomach.

c/f :

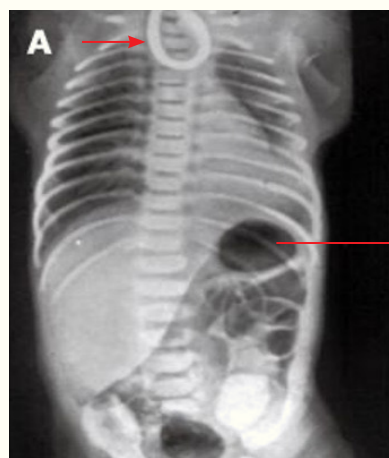
- Antenatal : Polyhydramnios.
- Postnatal :
 - Non-bilious vomiting (Onset : Immediately after birth).
 - Copious salivation.
 - **Aspiration** of feeds → Pneumonia.

Test :

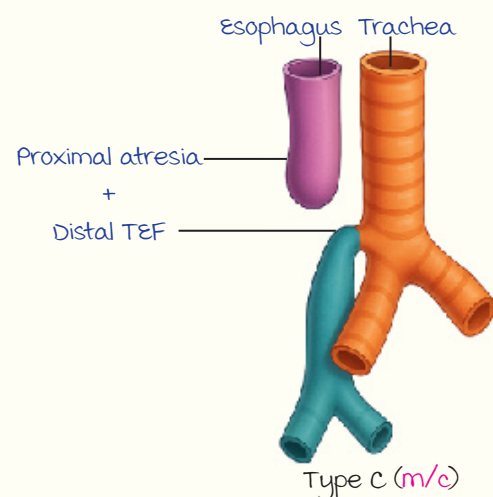
Passing of nasogastric (NG) tube (To check patency) → Resistance ⊕.

mx :

- NG tube : Drain proximal esophageal pouch.
- Sx : Ligation of TEF + anastomoses of esophageal ends.



Coiling of NG tube in proximal esophagus



Infantile GERD :

Risk factors :

- Large volume feeds.
- Supine posture.

Note :

Physiological gastroesophageal reflux (Periodic) :

In infants d/t

- Limited stomach capacity.
- Short esophageal length.

C/f : Recurrent non-bilious vomiting

+

- Failure to thrive.
- Aspiration pneumonia.
- Sandifer syndrome :
Severe GERD → Dystonic posturing of neck & back
↓
Relieve symptoms.

----- Active space -----



Sandifer syndrome

Course & mx :

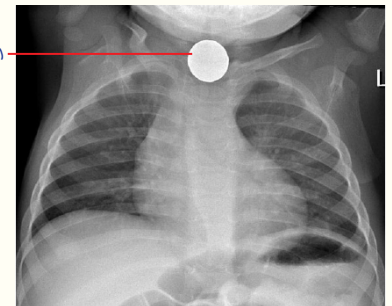
- Spontaneous resolution by 12 months.
- Feeds : Small, thick, frequent.
- Drugs : Proton pump inhibitors (PPI) x 3 months.
↓
No response
- Sx : Nissen's fundoplication.

Foreign Body in Esophagus 🐘 :

X-ray :

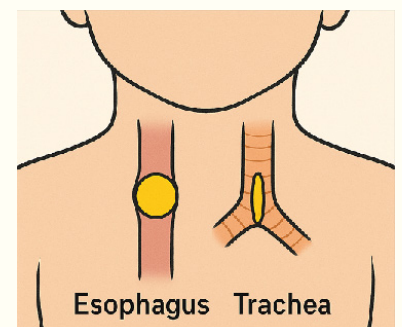
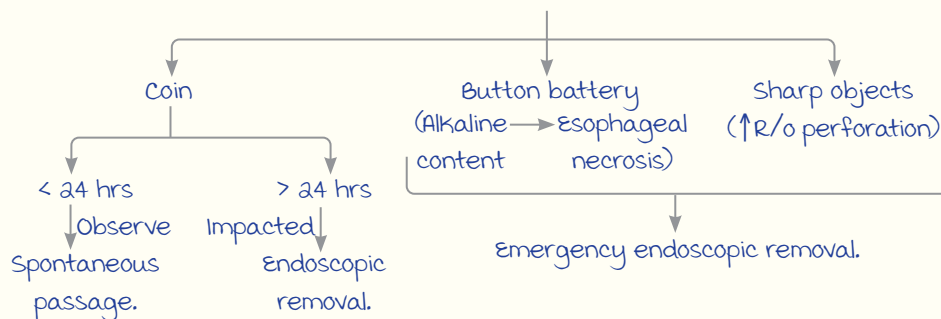
- Objects → Coin : Homogeneous opacity.
→ Button battery : Double halo sign.
- AP view → Esophagus : Flat, round.
→ Trachea : Slit-like.

Foreign body in esophagus



X-ray AP view

mx :



Coin in esophagus vs. trachea

Idiopathic Hypertrophic Pyloric Stenosis (IHPS) 🐘

00:16:25

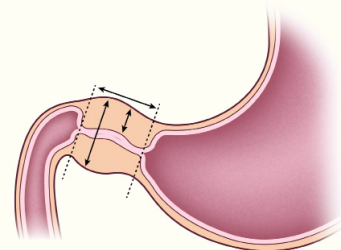
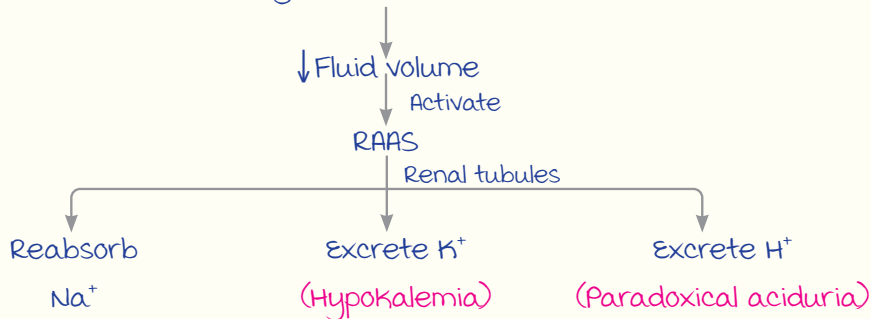
Pathophysiology :

Pylorus smooth muscles → Circular $\xrightarrow{\text{Hypertrophy}}$ Stenosis.
→ Longitudinal.

Associations : Erythromycin (macrolide).

----- Active space -----

Electrolyte abnormalities :

vomiting : Loss of fluids ($\downarrow \text{Na}^+$) + HCl (metabolic alkalosis, hypochloremia)

Pyloric stenosis

Clinical Features :

- Onset : 2 - 3 weeks after birth.
- Vomiting : Non-bilious, projectile.
- Olive - shaped epigastric mass.
- Visible gastric peristalsis.

Diagnosis :

USG : IOC.

Criteria : Pyloric

- muscle thickness : ≥ 4 mm.
- Channel length : ≥ 16 mm.

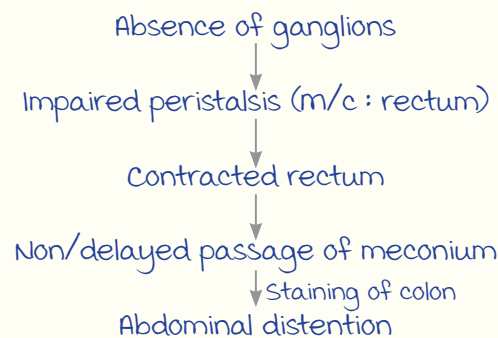
mx :

- Correct dehydration : NS + K^+ (IV fluid of choice).
- Definitive : Sx → Ramstedt's pyloromyotomy.

Hirschsprung Disease

00:23:50

AKA Aganglionosis.

Pathophysiology :Ix :

Rectal biopsy : IOC.

New stain : Calretinin.

Ganglions : Absent

↓ Loss of inhibition

Nerve fibres : Hypertrophy (Special stain : Acetylcholinesterase).

mx :

----- Active space -----

- Temporary : Diverting colostomy
- Definitive : Rectal resection + pull - through (Anastomosis of remaining colon).

Celiac Disease

00:28:05

Pathophysiology 🐘 :

Genetic predisposition : HLA DQ2, DQ8.

Gluten hypersensitivity : (Barley, rye, wheat)

T-cell Activate → B-cells → Autoantibodies

malabsorption ← Damage to small intestine



Dermatitis herpetiformis :

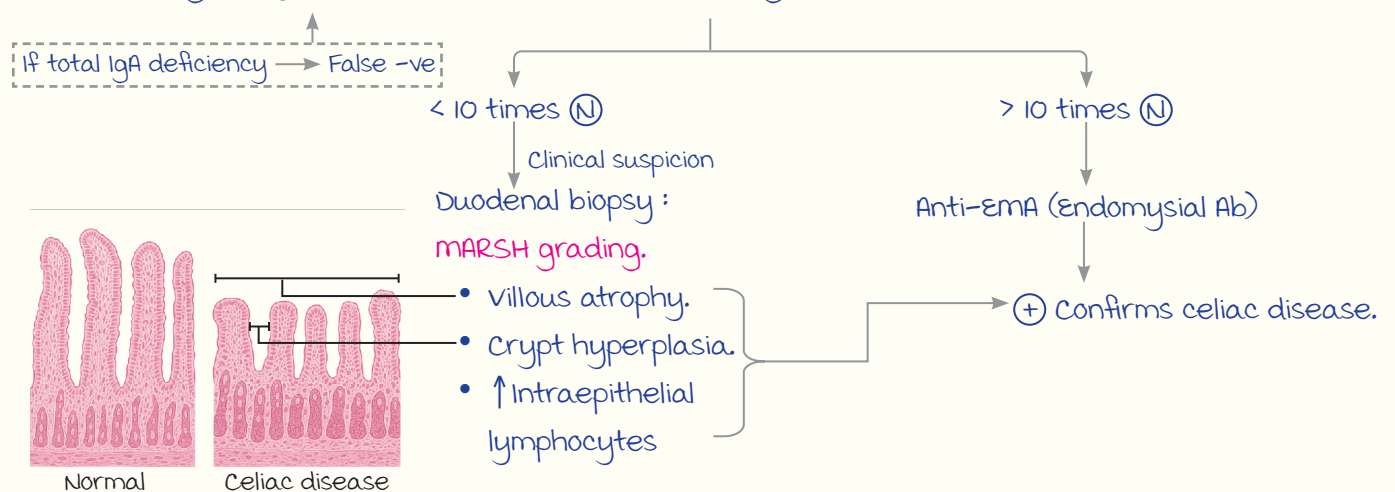
- Adults (m/c) : A/w lesions.
- Papulovesicular lesions.
- Predominant around joints (Extensors).

c/f :

- Chronic diarrhea.
- Failure to thrive.
- Anemia.

Diagnosis 🐘 :

Serum total IgA (major Ab) + Anti-tTG (Anti-tissue transglutaminase Ab : Initial Ab)

mx :

Gluten free diet : Lifelong.

----- Active space -----

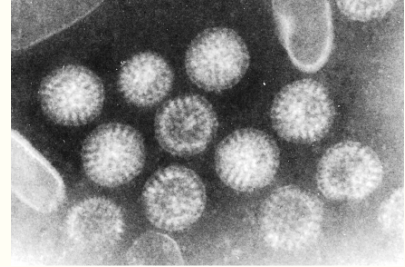
Acute Diarrhoea

00:36:00

- Duration : < 7 days.
- 2nd m/c cause of under-5 mortality.

Etiology :

m/c → Overall : **Rotavirus**.
 → Bacterial : **Enterotoxigenic E. coli (ETEC)**.



Em : Spoke wheel appearance of Rotavirus.

**Note :**

- m/c cause of under-5 mortality : Pneumonia.
- **Shigella flexneri** : m/c cause of dysentery.

Assessment & mx of Dehydration :

IMNCI guidelines :

	No dehydration	Some dehydration	Severe dehydration
Features	Active, alert child	Irritability, thirsty . (↑ water intake)	Lethargic
Skin pinch 	Fast response (< 1 sec)	Slow response (≤ 2 sec)	Very slow (> 2 sec)
management	Plan A : Replace ongoing loss per loose stools with ORS : • < 2 yrs : 50 – 100 mL • 2 – 10 yrs : 100 – 200 mL • > 10 yrs : As much as required	Plan B : 1. Rehydration with ORS : 75 mL/kg over 4 hrs 2. Replace ongoing loss 3. Daily/maintenance fluid requirements : Holliday Segar formula (Per day) : • < 10 kg : 100 mL/kg • 10 – 20 kg : 1000 mL + 50 mL/kg (For every kg > 10kg) • > 20 kg : 1500 mL + 20mL/kg (For every kg > 20kg)	Plan C 1. Rehydration with IV fluids : RL + 5% dextrose at 100 mL/kg • < 1yr : Over 6 hrs • > 1yr : Over 3 hrs

WHO low osmolar ORS :

Transport by SGLT-1 →

Components	Osm (mOsm/L)
Glucose	75
Na	75
K	20
Cl	65
Citrate	10
Total	245

Note :

Skin pinch not reliable in malnutrition (D/t loss of subcutaneous fat).

SYSTEMIC PAEDIATRICS : NEUROLOGY

Neural Tube Defects (NTD)

00:00:26

Pathophysiology :

Normal closure of neuropore
in 4th week of gestation

Failure →

NTD

Anterior → Cranial NTD.

Posterior (m/c) → Caudal NTD.

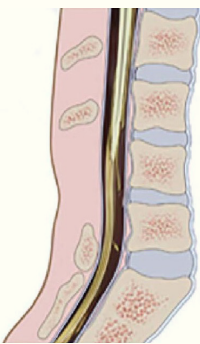
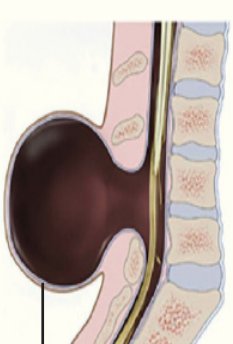
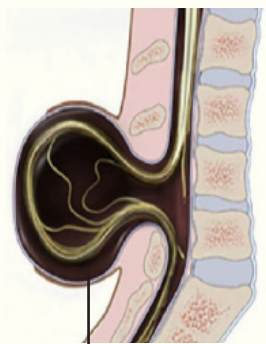
Risk Factors :

- Overt diabetes in mother
- Teratogenic drugs :
valproate, phenytoin, carbamazepine
- Folic acid deficiency

} multifactorial inheritance.

Defects :

Caudal NTD : m/c lumbosacral vertebrae (L₅/S₁).

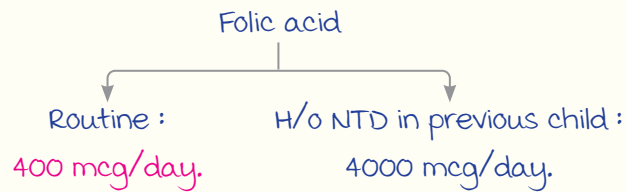
Spina bifida occulta	meningocoele	meningomyelocoele
		
Failure of fusion of posterior vertebral arch	Protrusion of meninges	Protrusion of meninges + Spinal nerve roots

Cranial NTD :

Anencephaly	Encephalocele	Craniorachischisis
		
• Absent skull • Absent cerebral hemispheres • Intact brainstem (Breathing & brainstem reflexes +ve)	• Skull defect → Herniation of brain tissue • m/c : Occipital area	• Spine & skull defect

Investigations :

- USG : 14 - 16 weeks POG.
- markers :
 - AFP (Screening).
 - **Acetylcholinesterase** (Best).

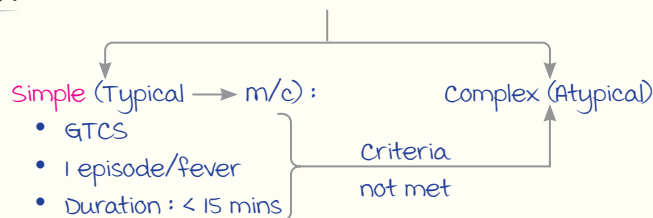
Prevention 

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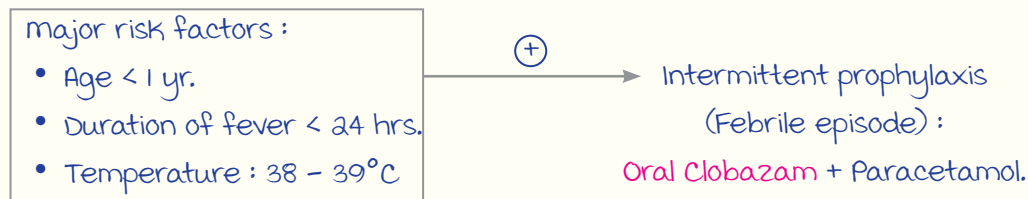
Febrile Seizure

00:10:04

- Overall m/c/c of seizure in children.
- Age : 6 months - 6 years.
- Previously (N) + fever + seizure.

Types :mx :

1. Stabilize : Airway, breathing, circulation.
2. Anti-pyretics, tepid sponging.
3. Look for reversible causes : Hypoglycemia → IV Dextrose.
4. Antiepileptics :
 - 1st line : **IV benzodiazepines**.
 - 2nd line : Fosphenytoin → Valproate → Levetiracetam.

Recurrence Risk of recurrence of subsequent epilepsy :

Risk factors	Risk stratification
• Simple febrile seizure • Recurrent febrile seizure	Low risk
• Complex febrile seizure • Family h/o epilepsy • Neurodevelopmental abnormalities	High risk : 1x to be done • EEG • Brain imaging

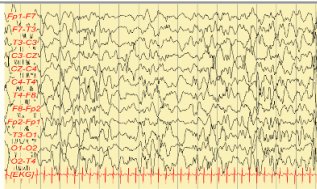
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Risk of recurrence :

- After 1 episode : 30%.
 - ≥ 2 episodes
 - Age < 1 year
- } 50%.

Epilepsy Syndromes

00:19:02

	Salient features	EEG	Rx
Juvenile myoclonic epilepsy/ Janz syndrome 	• Adolescents • On awakening (Sleep deprivation $\oplus$) • Gene : - GABRA-1 - CLCN2 - myoclonin-1	• 4 - 6 Hz • Polyspike $\&$ wave	valproate
Lennox Gastaut syndrome	• 2 - 10 years • Multiple seizure types • Developmental delay	• 1 - 2 Hz • Spike $\&$ slow wave	• Valproate • Rufinamide • Cannabidiol (CBD)
West syndrome/ Infantile spasm	• Age < 1 year. • Jack-knife seizure • Salaam spells • Developmental regression • Types : $\downarrow$ $\frac{1}{3}$ Cryptogenic/ idiopathic $\downarrow$ $\frac{2}{3}$ Symptomatic : - CNS defects - Worse prognosis	 Hypsarrhythmia : High voltage chaotic activity with multifocal spike $\&$ wave	• Inj. ACTH (1st line $\pm$ steroids) • Vigabatrin : - 2nd line - 1st line in : Tuberous sclerosis + West syndrome

Meningitis

00:26:50

Fever + signs of meningeal irritation.

Signs of meningeal Irritation :

uncommon in young children (Especially infants).

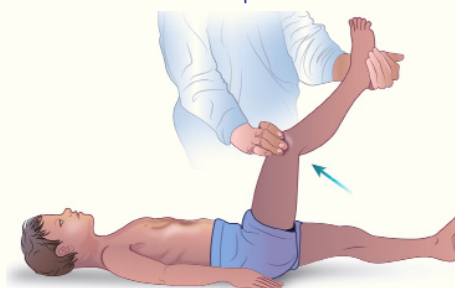
1. Nuchal rigidity : Neck stiffness.



3. Brudzinski's sign :

Flexion of neck $\rightarrow$ Flexion of leg.2. Kernig's sign :
Resistance to extension of leg

Back pain.



Etiology :

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Bacteria (m/c) :

- Overall m/c : *Streptococcus pneumoniae*.
- Neonates and first 2 months : *E. coli*.
- 2 - 24 months : *H. influenza* type B.

Virus :

Japanese encephalitis (m/c).

Investigation :

CSF analysis by lumbar puncture : IOC.

	Bacterial meningitis	Viral meningitis	TB meningitis (Cobweb appearance)
Sugar	Low	Normal	-
Protein	High	Normal / ↑	↑↑↑
WBC	↑: Neutrophils ++	↑: Lymphocytes ++	↑: Lymphocytes

Treatment :

1. IV antibiotics : 3rd gen. cephalosporins + vancomycin x 2 weeks.
2. IV steroids : Dexamethasone x 2 days.
 - ↓ Inflammation.
 - ↓ Incidence of sensorineural hearing loss (SNHL) : m/c long term sequelae.

Shunt Associated meningitis :

- H/o shunt for hydrocephalus.
- Organism : *S. epidermidis* (CONS : Coagulase negative staphylococcus).

mx : Shunt tap + CSF analysis

↓
vancomycin (x 14 days) + shunt removal & replace.meningococcal meningitis :2nd m/c/c.

Droplet precaution : Isolation until 24 hrs of starting antibiotic.

Chemoprophylaxis of contacts : Rifampicin x 2 days.

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Cerebral Palsy

00:36:30

- Birth asphyxia
 - Teratogen
 - TORCH infection
- Permanent damage to developing brain
- ↓
- Static encephalopathy (Not progressive).
 - Global developmental delay.
 - Low IQ, microcephaly.
 - Seizures.
 - Vision and hearing defects.



Scissoring posture :

- Tiptoe walking
- D/t ↑ tone in lower limb muscles

Types :

	Clinical features	Antecedent cause	Imaging
Spastic quadriplegia	All 4 limbs spasticity	Birth asphyxia in term	multicystic encephalomalacia
Spastic diplegia	Lower limbs spasticity : Scissoring posture	Birth asphyxia in preterm	Periventricular leukomalacia
Spastic hemiplegia	1/L upper & lower limb spasticity	Perinatal ischemic stroke	• Focal findings • Porencephalic cysts
Extrapyramidal	• Chorea • Athetosis	Bilirubin encephalopathy	Basal ganglia abnormalities
Ataxic	-	Cerebellar malformations	

Duchenne Muscular Dystrophy (DMD)

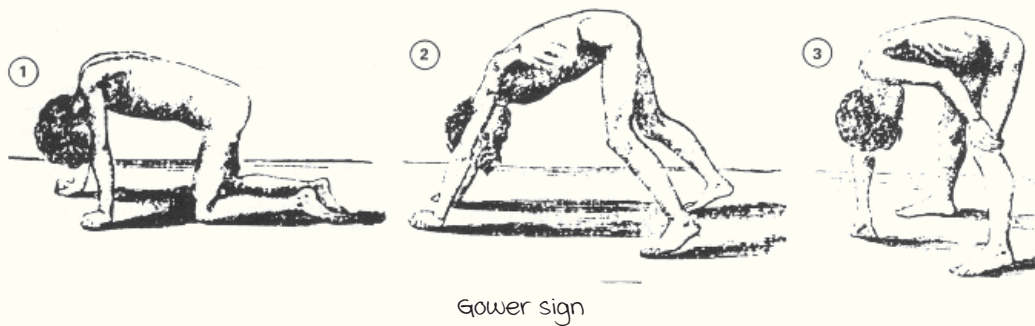
00:43:30

- X-linked recessive.
- Defect : Dystrophin gene → ↓ Dystrophin gene → Necrosis + fibro-fatty deposition in muscles.
(Frame-shift mutation)

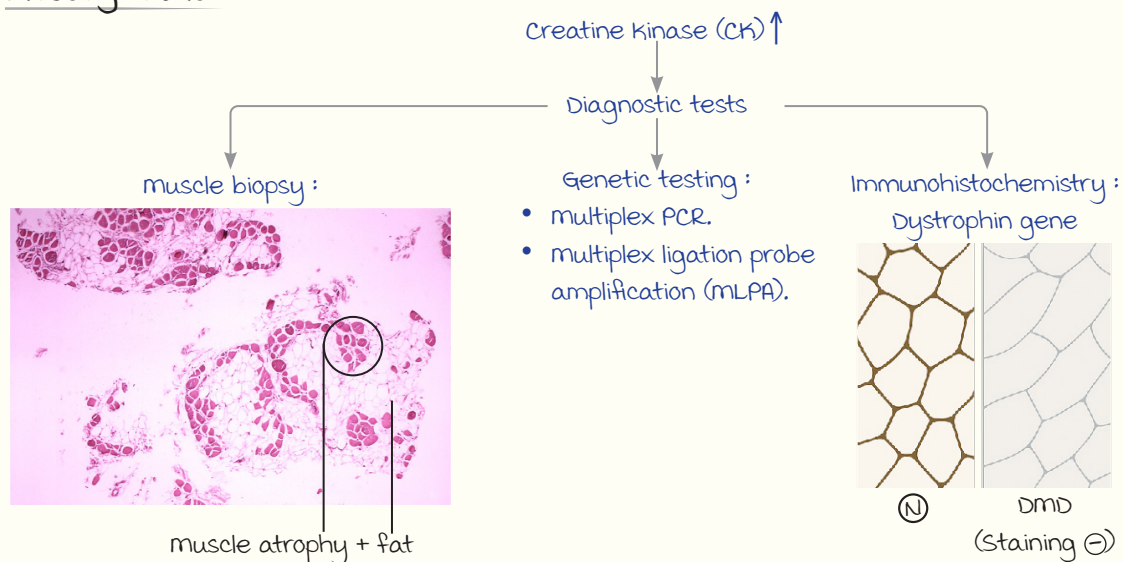
Clinical Features :

- Proximal muscle weakness (UL → Shoulder muscles, LL → Hip muscles) :
 - Delayed walking.
 - Difficulty in standing from sitting position : Elicited by Gower sign.
- Pseudohypertrophy :
 - D/t fibrofatty deposition.
 - Calf muscle (m/c site).
- Low IQ (< 70).
- Progressive scoliosis → Loss of ambulation (8½ - 10½ years).
- Cardiac failure (Cardiomyopathy).
- Respiratory failure : m/c cause of death.
- Death occurs at 20 - 25 years.

----- Active space -----



Pseudohypertrophy of calf muscles

Investigations :Miscellaneous Conditions

00:51:52

Becker's muscular Dystrophy :

- DMD variant.
- Inframe mutation of dystrophin gene : less severe.

Clinical :

- Onset : > 6 years.
- Slow progression.
- ↑ Risk of myalgia.
- Early cardiomyopathy.

viral myositis 🐘 :

- Influenza virus (m/c).
- within 1 week of infection.
- muscle pain.
- CK : ↑↑.
- self-limiting.

SYSTEMIC PAEDIATRICS : NEPHROLOGY

Congenital Abnormalities

00:00:10

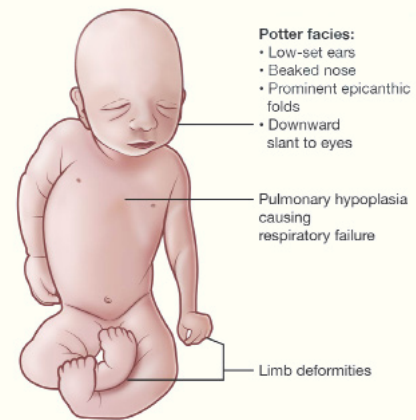
Renal Agenesis :

- W/L : Asymptomatic (Avoid contact sports)
- B/L : Antenatal ↓↓ urine output

Severe oligohydramnios

Fetal compression : **Potter's syndrome**

- Downslant eyes.
- Flat nose
- Lowset ears
- Retrognathia
- Skin wrinkles
- **Pulmonary hypoplasia** (m/c/c of death)
- Limb deformities



Posterior Urethral Valve :

- male baby.
- **m/c/c of obstructive nephropathy.**

Clinical features :

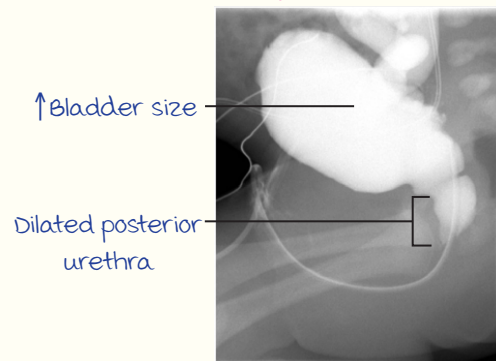
Antenatal :

- Hydroureteronephrosis
- Oligohydramnios

Postnatal :

- Poor urine stream
- Recurrent UTI.

IOC : **micturating cystourethrogram.**



Puv in MCU

Rx : Endoscopic fulguration of valve.

UTI

00:07:22

Etiology : **E. coli** (m/c).

Investigation :

urine culture : Diagnostic.

- Specimen : mid-stream clean catch urine sample.
- Dx : $\geq 10^5$ CFU/mL (CFU : Colony Forming unit).

Rx :

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Cystitis :

Oral antibiotics :

- Cefixime
- Coamoxiclav

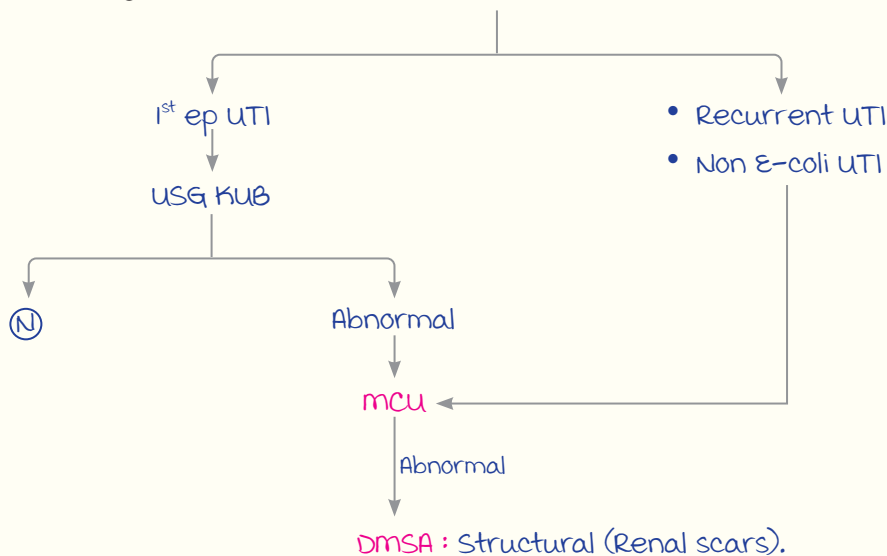
Pyelonephritis :

- Age < 3 months
- Sepsis
- Poor oral intake

↓ Yes

IV antibiotics :

- Cefotaxime (3rd gen cephalosporin).
- Gentamicin (Aminoglycoside).

Follow Up Investigation :Nephrotic Syndrome

00:12:30

Features :

- massive proteinuria :
 - urine protein : > 40 mg/m²/hr or > 1 g/m²/day.
 - Dipstick : + 3/ + 4.
 - P : C ratio : > 2.
- Hypoalbuminemia : < 3 g/dL.
- Generalized edema (Anasarca).
- Hyperlipidemia (↑ Lipoprotein).

Causes :

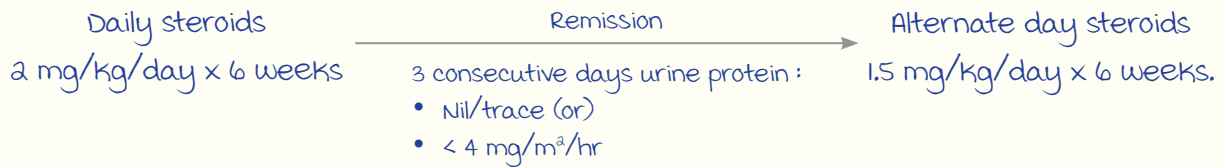
- minimal change disease (m/c) : Effacement of foot processes of podocytes
→ Proteinuria.
 - FSGS
 - membranous nephropathy
 - membranoproliferative nephropathy
- } Significant change nephrotic syndrome.

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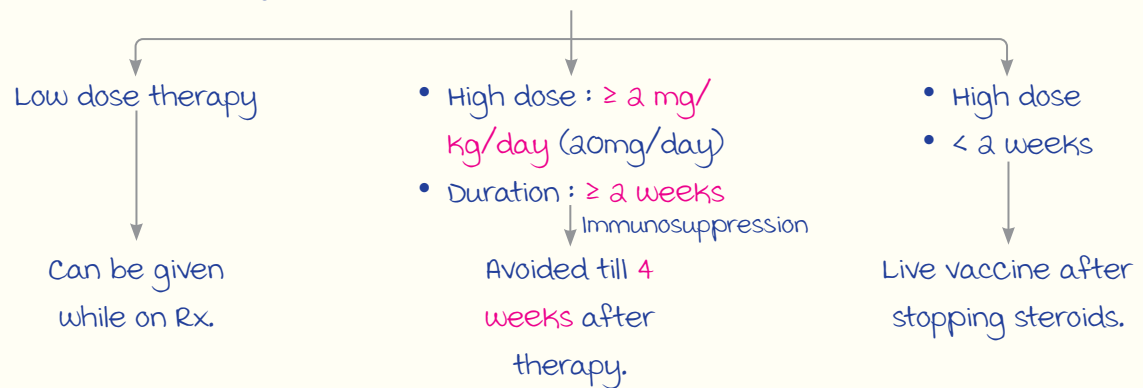
	minimal change disease	Significant change disorders (Eg : FSGS)
Age at onset	2-6 years	Older child
Hematuria/hypertension	⊖	±
Renal function	Normal	Abnormal/↓↓ leads to CKI
Response to steroids	Excellent	Resistant

Rx :

Steroids : Prednisolone.

**Note :**

Live vaccine during steroid Rx :

**Glomerulonephritis**

00:22:32

- Hematuria
 - Dysmorphic RBC & RBC casts.
 - Cola coloured urine.
- Hypertension.
- Peri-orbital puffiness.
- mild proteinuria

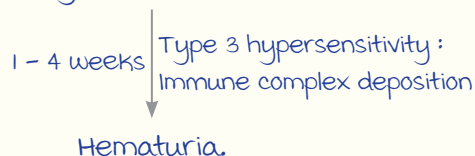
Post Streptococcal Glomerulonephritis (PSGN) 🐘 :

Age : 5 - 15 years.

Etiology :

Group - A β hemolytic streptococci :

- Pharyngitis : 4, 12 strain.
- Pyoderma : 49 strain.



Investigation :

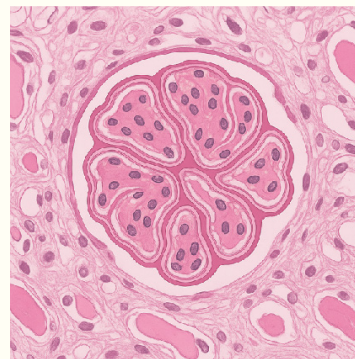
- $\uparrow$ ASO titre/ $\uparrow$ Anti-DNAse B.
- $\downarrow$ Serum C3 : Normalise by 8 - 12 weeks.

Treatment : Supportive.

Other Forms of Nephritis :

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	Features	Findings	Rx
Henoch-Schonlein purpura (HSP) nephritis	• m/c/c of vasculitis in children (4 - 8 yrs) • Small vessel vasculitis • IgA deposition	• Non-thrombotic palpable purpura (Lower limbs) • Abdominal pain • Arthralgia • Nephritis (1/3rd) : Hematuria/proteinuria Triad	• Self limiting • Steroid : Indication : - Severe GI pain/bleeding - RPGN
IgA nephropathy/ Berger's disease	1 - 2 days after viral URTI	• Recurrent gross hematuria • Biopsy : - mesangial proliferation - IgA deposits	ACE inhibitors
Lupus nephritis	> 50% of SLE	• Wide spectrum • Biopsy (Diagnosis & prognosis) : Wire loop lesions	• Induction phase (For 6 months) : - Cyclophosphamide (IV) + - Steroids • maintenance : mycophenolate



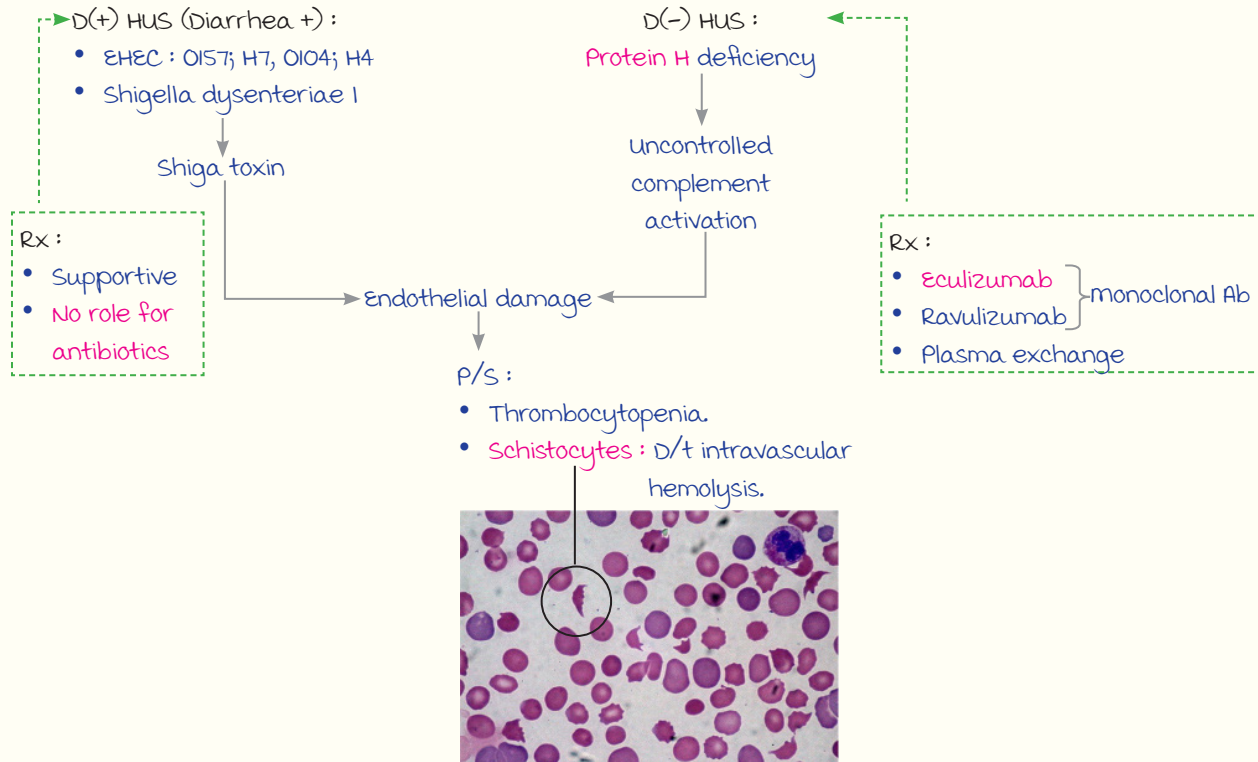
Wire-loop lesions :
Subendothelial immune deposits

----- Active space -----

Hemolytic Uremic Syndrome

00:37:10

Post infection acute kidney injury.

**KDIGO Criteria**

00:42:10

Stage of AKI	S. creatinine	urine output
1	↑ 1.5 - 2 fold	< 0.5 mL/kg/hr > 6 hrs
2	↑ > 2 - 3 fold	< 0.5 mL/kg/hr > 12 hrs
3	↑ > 3 fold (or) ≥ 4 mg/dL	< 0.3 mL/kg/hr for 24 hrs (or) anuria > 12 hrs

KDIGO : Kidney disease : Improving global outcomes.

Note :

$$\text{Estimated GFR (eGFR)} = \frac{\text{Length} \times \text{constant (K)}}{\text{S. creatinine}} \quad K = 0.41 - 0.43$$

SYSTEMIC PAEDIATRICS : ENDOCRINOLOGY

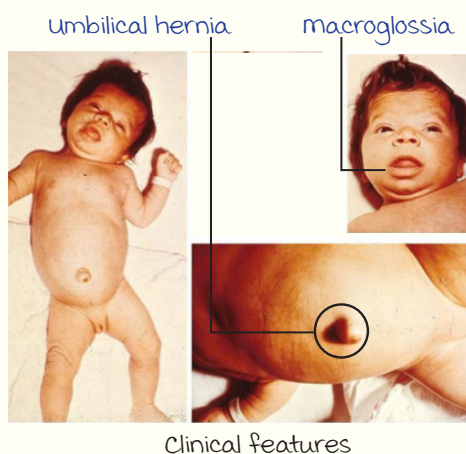
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Congenital Hypothyroidism

00:00:11

Etiology :

1. Thyroid dysgenesis (m/c) :
 - Thyroid hypoplasia.
 - Ectopic thyroid.
2. Dyshormonogenesis :
 - Defective thyroid hormone synthesis.
3. Iodine deficiency :
 - Endemic cretinism.
4. Iodine transport defects.
5. Maternal antibodies :
 - TSH receptor $\ominus$ antibodies.



Clinical Features :

1. General :
 - ↓ Activity.
 - Hoarse cry.
2. Skin :
 - Hypothermia.
 - Dry skin.
 - Prolonged jaundice (↓↓ UDP-GT activity → ↓ bilirubin conjugation).
3. GIT :
 - Constipation.
 - Delayed meconium passage after birth.
4. Fontanelles :
 - Wide open with delayed closure.

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Investigations :

1. Screening :

Dried blood spot (DBS) sample 2 - 4 days after birth
(False $\uparrow$ TSH in the first 2 days)

Serum TSH

>40 mIU/L

- Send TFT (FT_4 , T_3 , TSH)
- Start Rx : **Levothyroxine**

2. Radiology :

To finding etiology

Radionuclide scan :

$I^{123} > Tc^{99}$

USG thyroid.

Ectopic uptake

Ectopic thyroid.

No uptake

USG thyroid

Gland not visualised

Thyroid agenesis.

$\uparrow$ Uptake

Dyshormonogenesis
(D/t $\uparrow$ TSH).

Visualised

- **Iodine transport defects.**
- maternal antibodies.

Complications :

- Intellectual disability.
- Short stature.

Congenital Adrenal Hyperplasia (CAH)

00:09:10

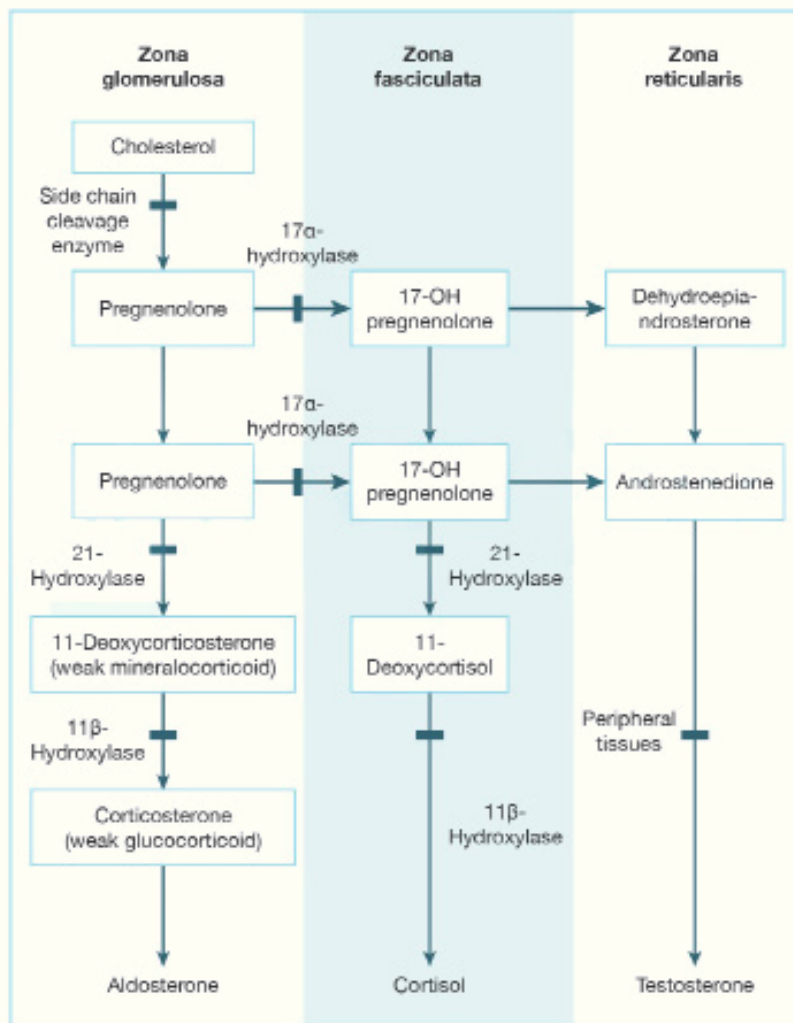
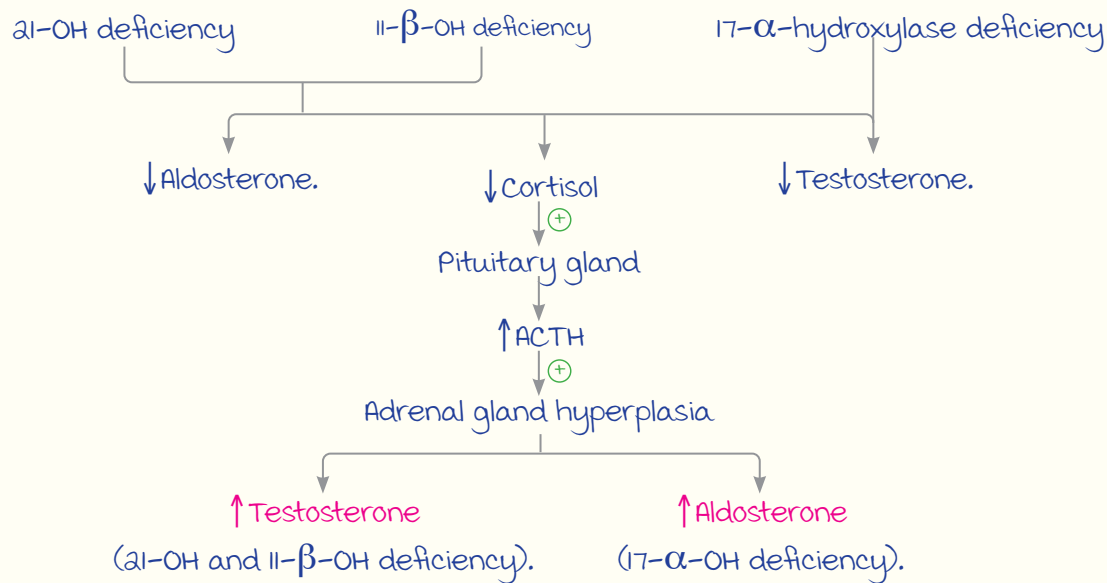
Etiology :

Deficiency of :

1. 21-hydroxylase (21-OH) : **m/c.**
2. 11- β -hydroxylase (11- β -OH).
3. 17- α -hydroxylase (17- α -OH).

Pathogenesis :

----- Active space -----



Pathway of adrenal steroid synthesis

----- Active space -----

Clinical Features :

Enzyme deficiency	Aldosterone		Cortisol		Testosterone	
	↑/↓	c/f	↑/↓	c/f	↑/↓	c/f
21-hydroxylase	↓	- Hypotension/shock - Hyponatremia - Hyperkalemia	↓	Hypoglycemia	↑	- Precocious puberty (♂) - Ambiguous genitalia (♀) (D/t virilization ↓ Clitoromegaly)
11-β-hydroxylase	↓	↑BP : Due to ↑11-deoxycorticosterone	↓		↑	- Precocious puberty (♂) - Ambiguous genitalia (♀)
17-α-hydroxylase	↑	↑BP	↓		↓	♂ : - Delayed puberty - Undervirilization - Ambiguous genitalia (Severe deficiency)

management of 21-OH Deficiency :

Screening marker : ↑17-OH progesterone.

Treatment :

- Fludrocortisone
 - Hydrocortisone
- } Lifelong.

Prevention :

Dexamethasone antenatally : ↓Risk of virilization (Due to CAH) in female fetus.